MLHVDC-LCS-ECI-DD-HA-0101-001A

30-B003701S-N0101-001

巴基斯坦默蒂亚里—拉合尔±660kV 直流输电工程
拉合尔±660kV 换流站工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

±660kV Lahore Converter Station

施工图设计
Detail Design

暖 通 部 分

HVAC Part

暖通设计说明

Design Specification of HVAC System

a) 中国电力技术装备有限公司
China Electric Power Equipment and Technology Co., Ltd.
b) 中国电力工程顾问集团华东电力设计院有限公司
East China Electric Power Design Institute of China Power Engineering Consulting Group
April, 2019

拉合尔±660kV 换流站 工程

施工图 阶段

	中国电力工程顾问集团华东电力设计院有限公司 工程设计综合资质甲级 A131000025 工程勘察综合类甲级 B131000025

卷册检索号		30	B003701S		N0101			
文本编号		B003701S		N0101			001	
0	201904	CFC	新制					
版号	日期	状态	修改内容摘要	设计	校核	审核		批准

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任。

批准		2019.04	暖通设计说明 Design Specification of HVAC System	
审核		2019.04		
校核		2019.04		
设计		2019.04		
日期	2019.04	状态	CFC	版号	A

目 录

Table of Contents

1 概述 1
1 Overview 1
1.1 设计依据 1
1.1 Design basis 1
1.2 工程设计概况 1
1.2 Overview of Engineering Design 1
1.3 设计范围 1
1.3 Design Scope 1
1.4 设计原始资料 2
1.4 Original design data 2
2 主要设计原则及标准 3
2 Main design principles and standards 3
2.1 主要设计原则 3
2.1 Main design principles 3
2.2 设计标准 7
2.2 Main design specification 7
3 系统设计方案及主要技术说明 7
3 System design scheme and main technical description 7
3.1 空调冷热源系统 7
3.1 Cooling and heat source system 7
3.2 阀厅通风和空调 12
3.2 Ventilation and air conditioning of Valve hall 12
3.3 主控楼通风和空调 18
3.3 Ventilation and air conditioning of centralized control building 19
3.4 #1 继电器室及 11kV 站用电室通风及空调 37

3.4 Ventilation and air conditioning of #1 Relay and 11kV Station Service House.....37
3.5 #2 继电器室及计量小室通风及空调.....38
3.5 Ventilation and air conditioning of #2 Relay and Metering House.....38
3.6 #3 继电器室通风及空调.....40
3.6 Ventilation and air conditioning of #3 Relay House.....40
3.7 400V 站用电室通风及空调.....41
3.7 Ventilation and air conditioning of 400V Station Service House.....41
3.8 综合泵房及消防泵房通风及空调.....42
3.8 Ventilation and air conditioning of Pump House and Fire Pump House.....42
3.9 专用品库空调.....43
3.9 Air conditioning of Special materials storage.....43
3.10 深井泵房通风.....44
3.10 Ventilation of Deep well.....44
3.11 站前区办公楼通风及空调.....44
3.11 Ventilation and air conditioning of Office building in the first building area.....44
3.12 站前区中方宿舍通风及空调.....45
3.12 Ventilation and air conditioning of Domitory building for chinese in the first building area.....45
3.13 中方食堂通风及空调.....46
3.13 Ventilation and air conditioning of Refectory for chinese.....46
3.14 活动室通风及空调.....47
3.14 Ventilation and air conditioning of Activity house.....47
3.15 站前区巴方宿舍通风及空调.....47
3.15 Ventilation and air conditioning of Domitory building for pakistani in the first building area.....47
3.16 巴方食堂通风及空调.....48
3.16 Ventilation and air conditioning of Refectory for pakistani.....48
3.17 门卫室通风及空调.....48

3.17 Ventilation and air conditioning of Guard room 49
3.18 建维区办公楼通风及空调 49
3.18 Ventilation and air conditioning of Office building in the temporary building area 49
3.19 管理人员宿舍通风及空调 50
3.19 Ventilation and air conditioning of Domitory building for managers 50
3.20 员工宿舍通风及空调 51
3.20 Ventilation and air conditioning of Domitory building for staff 51
3.21 巴方安保宿舍通风及空调 51
3.21 Ventilation and air conditioning of Domitory building for security 51
3.22 防火防爆 52
3.22 Fire proof and explosion proof 52
4 暖通专业施工图卷册目录 54
4 Drawing list of HVAC Part 54

1 概述

1 Overview

1.1 设计依据

1.1 Design basis

本工程施工图设计阶段主要依据文件如下：

- 我院编制的拉合尔±660kV换流站工程初步设计文件；
- 工程相关会议纪要；
- 国家及行业现行的标准、规程、规范、技术条例及强制性标准。

The main documents for the detail design stage of the construction drawings are as follows:

- The preliminary design document of Lahore (+660 kV) converter station project compiled by our institute;
- Summary of engineering related meetings;
- Current national and industry standards, regulations, norms, technical regulations and mandatory standards.

1.2 工程设计概况

1.2 Overview of Engineering Design

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程起点为信德省默蒂亚里，落点旁遮普省拉合尔，线路长度约878km，规划输电规模4000MW，由中国电力技术装备有限公司（CET）以BOOT模式开展。拉合尔换流站工程额定直流电压为±660kV，额定直流电流3030A，输送容量4000MW。

The starting point of Pakistan's Mertiary-Lahore (+660 kV) HVDC transmission project is from Mertiary in Sindh Province to Lahore in Punjab Province, with a transmission line length of about 878 km and a planned transmission scale of 4000MW. The project is carried out by China Electric Power Technology Equipment Co., Ltd (CET) in the BOOT mode. The rated DC voltage is ±660kV, rated DC current is 3030A, and transmission capacity is 4000MW.

1.3 设计范围

1.3 Design Scope

本设计范围为巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程受端拉合尔换流站内的各建筑物通风、空调系统的设计。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

This design scope covers the design of ventilation and air conditioning systems for buildings in Lahore Converter Station of Mutiari-Lahore (+660 kV) HVDC transmission project in Pakistan.

1.4 设计原始资料

1.4 Original design data

1.4.1 工程气象资料

1.4.1 Meteorological parameters

拉合尔气象站常规气象资料如下（除个别值外，资料统计年限为 1980-2010）：

The conventional meteorological data of Lahore Station are as follows (except for individual values, the statistical years of data are 1980-2010):

年平均气温 Average annual temperature：23.9℃

极端最高气温 Extreme maximum temperature：48.0℃(2007.6.10)

极端最低气温 Extreme minimum temperature：-2.2℃(2008.1.22)

年平均气压 Annual average pressure：983.1hPa

年平均相对湿度 Average annual relative humidity：57%

年平均风速 Average annual wind speed：1.6m/s

实测最大风速 Maximum measured wind speed：40.7m/s（3s 瞬时）

平均降水量 Average precipitation ：679.7mm

24 小时最大降雨量 Maximum 24-hour rainfall：274mm（2014.9.6）

年平均蒸发量 Average annual evaporation：1623.6mm

主导风向 Leading wind direction：NW

1.4.2 室外设计参数

1.4.2 Outdoor design parameters

通风和空调室外设计参数相关数据来自《ASHRAE Handbook—Fundamentals》中的 LAHORE AIRPORT 数据。

Data related to outdoor design parameters of ventilation and air conditioning are from LAHORE AIRPORT data in ASHRAE Handbook-Fundamentals.

气象台站位置 Location of meteorological stations：31.52N，74.40E

供暖室外计算温度/ Heating DB：3.9℃（99.6%）

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

夏季室外计算干球温度/ Cooling DB：43.2℃（0.4%）

夏季室外计算湿球温度/ Cooling MCWB：23.2℃（0.4%）

2 主要设计原则及标准

2 Main design principles and standards

2.1 主要设计原则

2.1 Main design principles

（1）本站共 2 个阀厅，每个阀厅设计一套独立的全空气集中空调系统，采用风冷热泵机组＋组合式空气处理机组的系统型式。风冷热泵机组和组合式空气处理机组均一用一备。空调冷冻水设计供水温度为 7℃、回水温度为 12℃，空调热水设计供水温度为 45℃、回水温度为 40℃，风冷热泵、组合式空气处理机组及相关系统设备均布置对应阀厅室外地面。

(1) There are two valve halls in this station. Each valve hall set an independent all-air centralized air conditioning system, which consist of air-cooled heat pump unit + air handling unit (AHU). The air-cooled heat pump units and the AHUs are all 100% backup. The design chilled water supply temperature is 7℃, the chilled water return temperature is 12 ℃, the design heating water supply temperature is 45 ℃, and the heating water return temperature is 40 ℃. The air-cooled heat pump, AHU and related equipment are all arranged outside the corresponding valve hall.

（2）本站主控楼区域设计一套独立的空调冷热源系统，采用 2x100%的风冷热泵机组。空调冷冻水设计供水温度为 7℃、回水温度为 12℃，空调热水设计供水温度为 45℃、回水温度为 40℃，风冷热泵及相关的水系统设备均布置主控楼屋面。

（2）A set of independent air-conditioning cold and heat source system is designed in the main control building area of the station, with 2x100% air-cooled heat pump unit. The design chilled water supply temperature is 7℃, the chilled water return temperature is 12 ℃, the design heating water supply temperature is 45 ℃, and the heating water return temperature is 40 ℃. The air-cooled heat pump and related water system equipment are arranged on the roof of main control building.

（3）主控楼内换流变直流场接口设备室、阀冷控制保护设备室、控制保护设备室和主控室等房间设置全空气空调系统，相应系统的空气处理机组采用 2x100%容量配置。主控楼内礼拜室、资料室、休息室、办公室和会议室等房间设置一套风机盘管+新风的空调系统。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

（3）For Converter Transformer Room, DC Distribution Panel Room, Cooling Control Protection Equipment Room and main control room etc. in the main control building, the air handling unit of the corresponding system is equipped with 2x100% capacity. Fan coil unit and fresh air in the chapel room, reference room, lounge, offices and conference room are set.

（4）主控楼内交流阀组配电室、阀冷却设备间分别设置蒸发冷却系统，每套系统采用 2x100% 容量配置。

（4）In the main control building, evaporative cooling system is set up in the 380/220V AC transformer rooms and Cooling installation rooms. Each system is equipped with 2x100% capacity.

（5）蓄电池室设置防爆型热泵分体空调，另设置排风机作为事故时排风。

（5）Explosion-proof heat pump split air-conditioning is installed in the storage battery room, and exhaust fan is installed in case of accident.

（6）阀厅设灾后排烟系统，主控楼内各电气设备间设置独立的灭火后排风系统。

（6）The valve hall is equipped with post-disaster smoke exhaust system, and the main control building is equipped with independent ventilation system after extinguishing for the electrical equipment rooms.

（7）站内办公楼、食堂、宿舍楼等建筑物设置风冷热泵分体式空调。

（7）Air-cooled heat pump split air conditioner is installed in office buildings, canteens and dormitory buildings.

（8）站内其它辅助建筑各房间根据工艺及人员舒适要求设置通风或分体空调装置。

（8）All rooms of other auxiliary buildings in the station will be equipped with ventilation or split air conditioning devices according to the requirements of technology and personnel comfort.

（9）站内的集中冷冻水系统、集中空调系统、蒸发冷却通风系统将设置由微处理控制器控制的两级自动监控系统。

（9）The two-stage automatic monitoring system controlled by the microprocessor controller will be set up for the centralized chilled water system, the centralized air conditioning system, and the evaporative cooling ventilation system in the station.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

（10）各空调/通风系统均和消防控制连锁，火灾报警时联动关闭相应区域所有空调和风机设备。

（10）Each air conditioning/ventilation system is interlocked with fire control system, and all air conditioners and fan equipment in the corresponding area are closed in conjunction with the fire alarm.

（11）换流站各建筑物或房间室内设计参数及系统型式：

（11）Interior design parameters and system types of buildings or rooms in the station:

表 2.1-1 室内设计参数表

Table 2.1-1 Interior design parameter table

序号 No.	建筑物/房间名称 Building/Room Name	系统类型 System	室内温度 Indoor Temperature		室内相对湿度 Indoor Humidity		备注 Remark
			夏季 Summer	冬季 Winter	夏季 Summer	冬季 Winter	
1	**阀厅 Valve Hall**	空调+蒸发冷却 AC+EC	45°C	15°C	25～60%	25～60%	灾后排烟 Smoke exhaust
2	**主控楼 Main control building**						
2.1	主控室 Main Control Room	空调 AC	25～27°C	18°C	45～65%	45～65%	
2.2	控制保护设备室、直流屏室、通信机房等 Control Protection Equipment Room、DC Distribution Panel Room、Communication Room etc.	空调 AC	25～27°C	18°C	45～65%	45～65%	灭火后排风 Ventilation after extinguishing
2.3	380/220V 交流阀组配电室、阀冷却设备间 380/220V AC transformer room、Cooling Installation Room	蒸发冷却 EC	≤35	/	/	/	灭火后排风 Ventilation after extinguishing

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

序号 No.	建筑物/房间名称 Building/Room Name	系统类型 System	室内温度 Indoor Temperature		室内相对湿度 Indoor Humidity		备注 Remark
			夏季 Summer	冬季 Winter	夏季 Summer	冬季 Winter	
2.4	蓄电池室 Storage Battery Room	空调 AC	≤30	20	/	/	事故排风 Emergency ventilation
2.5	会议室、办公室等 Office etc.	空调 AC	26～28	18～20	/	/	
2.6	卫生间 Toilet	机械通风 Ventilation	/	/	/	/	
3	**站内其他建筑** **Other buildings**						
3.1	继电器室 Relay room	空调 AC	24～28°C	18～20°C	45～65%	45～65%	灭火后排风 Ventilation after extinguishing
3.2	蓄电池室 Storage Battery Room	空调 AC	≤30	20	/	/	事故排风 Emergency ventilation
3.3	站用电室 Station Service House	空调 AC	≤35	/	/	/	灭火后排风 Ventilation after extinguishing
3.4	水泵房 Pump room	机械通风 Ventilation	/	/	/	/	
3.5	专用品库 Materials storage room	空调 AC	24～28	18～20	/	/	/
3.6	食堂、宿舍楼、办公楼、活动室 Refectory, Domitory, Office and Activity house	空调 AC	26～28	18～20	/	/	

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

2.2 设计标准

2.2 Main design specification

本工程通风和空调系统的设计符合如下（但不限于此）现行的国家规范和标准：

The design of ventilation and air conditioning system in this project conforms to the following (but not limited to) current national standards:

《工业建筑供暖通风与空气调节设计规范》（GB50019-2015）

"Design code for heating ventilation and air conditioning of industrial buildings"(GB50019-2015)

《民用建筑供暖通风与空气调节设计规范》（GB50736-2012）

"Design code for heating ventilation and air conditioning of civil buildings"（GB50736-2012）

《高压直流换流站设计规范》（GB/T 51200-2016）

"Code for design of HVDC converter station"（GB/T 51200-2016）

《发电厂供暖通风与空气调节设计规范》（DL/T5035-2016）

"Code for design of heating ventilation and air conditioning of power plant"（DL/T5035－2016）

《建筑设计防火规范》（GB50016-2014）

"Code of design on building fire protection and prevention" (GB50016－2014)

《火力发电厂与变电站设计防火规范》（GB50229-2006）

"Code for design of fire protection for fossil fuel power plants and substations"（GB50229-2006）

3 系统设计方案及主要技术说明

3 System design scheme and main technical description

3.1 空调冷热源系统

3.1 Cooling and heat source system

本工程共设置 3 套集中空调冷热水系统，分别为极 1 阀厅、极 2 阀厅和主控楼提供空调冷冻水和热水。其中夏季冷冻水供回水温度为 7℃/12℃，冬季热水供回水温度为 45℃/40℃。

每套空调冷热水系统主要由 2x100%容量的风冷热泵机组、2x100%容量的冷热水循环泵、1 台自清洗水过滤器、1 台水处理器、1 台定压补水机组以及管道系统和电气配电及仪表控制系统组成。极 1 阀厅、极 2 阀厅空调冷热水系统相关设备布置于相应阀厅室外地面，

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

主控楼空调冷热水系统相关设备布置于主控楼 17.00m 屋面。

极 1 阀厅空调冷热水系统图纸见 30-B003701S-N0203 卷册。

极 2 阀厅空调冷热水系统图纸见 30-B003701S-N0204 卷册。

主控楼空调冷热水系统图纸见 30-B003701S-N0201 卷册。

Three sets of central cooling and heat source systems are set up in this project, which provide chilled water and hot water for pole 1 valve hall, pole 2 valve hall and main control building respectively. The supply and return temperature of chilled water in summer is 7/12℃, and temperature of hot water for supply and return in winter is 45/40℃.

Each set of cooling and heat source system is mainly composed of 2x100% capacity air-cooled heat pump unit, 2x100% capacity water circulating pump, 1 self-cleaning water filter, 1 water processor, 1 constant pressure make up water supply unit, pipeline system, electrical distribution and instrument control system. The related equipment of the cooling and heat source system of the pole 1 valve hall and the pole 2 valve hall are arranged outside the corresponding valve hall, and the related equipment of the cooling and heat source system of the main control building is arranged on the 17.00m roof of the main control building.

Cooling and heat source system for Pole 1 Valve hall refer to 30-B003701S-N0203.

Cooling and heat source system for Pole 2 Valve hall refer to 30-B003701S-N0204.

Cooling and heat source system for Main control building refer to 30-B003701S-N0201.

3.1.1 阀厅空调冷热水系统

3.1.1 Cooling and heat source system for Valve hall

(1) 极 1 阀厅和极 2 阀厅分别设置独立的空调系统，其空调冷热负荷如下表所示：

(1) The pole 1 valve hall and the pole 2 valve hall are equipped with independent air conditioning systems. The cooling and heating loads of air conditioning are shown in the following table:

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

表 3.1-1 阀厅空调冷热负荷

Table 3.1-1 Cooling and heating load of Valve hall

序号/No.	房间/Room	空调面积/AC area (m^2)	房间设计冷负荷/Design cooling load (kW)	房间设计热负荷/Design heating load (kW)	系统冷负荷/System cooling load (kW)	系统热负荷/System heating load (kW)
1	极 1 阀厅 Valve hall pole1	2975	294.5	119	475.2	128.3
2	极 2 阀厅 Valve hall pole1	2975	294.5	119	475.2	128.3

(2) 阀厅空调冷热水系统组成

(2) Form of Cooling and heat source system for Valve hall

极 1 和极 2 阀厅空调系统相同，单个阀厅空调冷热水系统主要由以下各部分组成：

- 2x100%容量的风冷热泵机组：LSRF500A，设计工况下，制冷量：500kW，制热量：550kW，制冷剂：R134a，制冷性能系数≥2.70。
- 2x100%容量的冷热水循环泵：GLC100-160，流量：94m³/h，扬程：352kPa。
- 1 台自清洗水过滤器：设计流量：94m³/h，阻力：0.01MPa。
- 1 台水处理器：激光负离子智能型水处理器，设计流量：94m³/h。
- 1 台定压补水机组：SXBY-600，膨胀罐 DN600，全容积：0.35m³，补水箱有效容积：1.25m³,补水泵 2 台，流量：2.5m³/h，扬程：200kPa，系统定压点压力为 0.10MPa。配套电控柜。
- 系统管路、阀门及电气控制装置等。

The air-conditioning system of pole 1 and pole 2 valve halls is the same. The cooling and heat source system of single valve hall consists of the following parts:

- 2 x 100% capacity air-cooled heat pump unit: LSRF500A, design conditions, Cooling capacity: 500 kW, Heating capacity: 550 kW, refrigerant: R134a, refrigeration coefficient of performance (> 2.70).
- 2x100% capacity water circulating pump: GLC100-160, flow rate: 94m^3/h, head: 352kPa.
- 1 self-cleaning water filter: design flow: 94m^3/h, resistance: 0.01MPa.
- 1 water treatment device: laser anion intelligent water treatment device, designed flow rate:

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

94m^3/h.

- 1 constant pressure make up water supply unit: SXBY-600, expansion tank DN600, full volume: 0.35m^3, effective volume of water supply tank: 1.25m^3, two water supply pumps, flow rate: 2.5m^3/h, head: 200kPa, system pressure at fixed point is 0.10MPa. Equipped with electronic control cabinet.
- Piping, valves and electrical control devices.

（3）空调冷热水系统运行和控制

(3) Operation and Control of cooling and heat source system

集中冷冻水系统夏季与过渡季运行，集中热水系统冬季运行。

系统启停顺序：启动膨胀补水定压机组，开启风冷热泵机组入口电动切换蝶阀，启动循环水泵，启动风冷热泵机组；停机顺序与启动顺序相反。

每套集中冷热水系统分别配置一套就地自动监控系统，就地监控系统均纳入暖通中央监控系统。自动监控系统对设备启停、联动、运行状态及运行时间，电动阀门开关及状态，水流量、供回水温度、压力，水过滤器压差等进行自动监控及保护。

The centralized chilled water system runs in summer and transition seasons, and the centralized hot water system runs in winter.

Start-stop sequence of the system: start the constant pressure make up water supply unit, open the electric switch butterfly valve at the inlet of the air-cooled heat pump unit, start the circulating water pump, start the air-cooled heat pump unit; stop sequence is opposite to start sequence.

Each cooling and heat source system is equipped with a local automatic monitoring system, which is integrated into the central heating and ventilation monitoring system. The automatic monitoring system automatically monitors and protects the start-stop, linkage, operation state and operation time of the equipment, the switch and state of the electric valve, water flow, temperature and pressure of water supply and return, pressure difference of water filter, etc.

3.1.2 主控楼空调冷热水系统

3.1.2 Cooling and heat source system for Main control building

(1) 主控楼空调冷热负荷如下表所示：

The cooling and heating loads of air conditioning are shown in the following table:

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

表 3.1-2 主控楼空调冷热负荷

Table 3.1-2 Cooling and heating load of Main control building

序号/No.	房间/Room	空调面积/AC area (m^2)	房间设计冷负荷/Design cooling load (kW)	房间设计热负荷/Design heating load (kW)	系统冷负荷/System cooling load (kW)	系统热负荷/System heating load (kW)
1	主控楼 Main control building	1342	288.3	95.8	410.2	145.3

（2）主控楼空调冷热水系统组成

(2) Form of Cooling and heat source system for Main control building

主控楼空调冷热水系统主要由以下各部分组成：

- 2x100%容量的风冷热泵机组：LSRF440A，设计工况下，制冷量：442kW，制热量：491kW，制冷剂：R134a，制冷性能系数≥1.96。
- 2x100%容量的冷热水循环泵：GLC80-200，流量：83m³/h，扬程：300kPa。
- 1 台自清洗水过滤器：设计流量：83m³/h，阻力：0.01MPa。
- 1 台水处理器：激光负离子智能型水处理器，设计流量：83m³/h。
- 1 台定压补水机组：SXBY-600，膨胀罐 DN600，全容积：0.35m³，补水箱有效容积：1.25m³,补水泵 2 台，流量：2.5m³/h，扬程：200kPa，系统定压点压力为 0.10MPa。配套电控柜。
- 系统管路、阀门及电气控制装置等。

The cooling and heat source system of main control building consists of the following parts:

- 2 x 100% capacity air-cooled heat pump unit: LSRF440A, design conditions, Cooling capacity: 442 kW, Heating capacity: 491 kW, refrigerant: R134a, refrigeration coefficient of performance (> 2.70).
- 2x100% capacity water circulating pump: GLC80-200, flow rate: 83m^3/h, head: 300kPa.
- 1 self-cleaning water filter: design flow: 83m^3/h, resistance: 0.01MPa.
- 1 water treatment device: laser anion intelligent water treatment device, designed flow rate: 83m^3/h.
- 1 constant pressure make up water supply unit: SXBY-600, expansion tank DN600, full volume: 0.35m^3, effective volume of water supply tank: 1.25m^3, two water supply pumps,

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

flow rate: 2.5m^3/h, head: 200kPa, system pressure at fixed point is 0.10MPa. Equipped with electronic control cabinet.

- Piping, valves and electrical control devices.

（3）空调冷热水系统运行和控制

(3) Operation and Control of cooling and heat source system

集中冷冻水系统夏季与过渡季运行，集中热水系统冬季运行。

系统启停顺序：启动膨胀补水定压机组，开启风冷热泵机组入口电动切换蝶阀，启动循环水泵，启动风冷热泵机组；停机顺序与启动顺序相反。

每套集中冷热水系统分别配置一套就地自动监控系统，就地监控系统均纳入暖通中央监控系统。自动监控系统对设备启停、联动、运行状态及运行时间，电动阀门开关及状态，水流量、供回水温度、压力，水过滤器压差等进行自动监控及保护。

The centralized chilled water system runs in summer and transition seasons, and the centralized hot water system runs in winter.

Start-stop sequence of the system: start the constant pressure make up water supply unit, open the electric switch butterfly valve at the inlet of the air-cooled heat pump unit, start the circulating water pump, start the air-cooled heat pump unit; stop sequence is opposite to start sequence.

Each cooling and heat source system is equipped with a local automatic monitoring system, which is integrated into the central heating and ventilation monitoring system. The automatic monitoring system automatically monitors and protects the start-stop, linkage, operation state and operation time of the equipment, the switch and state of the electric valve, water flow, temperature and pressure of water supply and return, pressure difference of water filter, etc.

3.2 阀厅通风和空调

3.2 Ventilation and air conditioning of Valve hall

极 1 阀厅和极 2 阀厅通风和空调系统相同，极 1 阀厅通风和空调系统图纸见 30-B003701S-N0203 卷册。极 2 阀厅通风和空调系统图纸见 30-B003701S-N0204 卷册。

The ventilation and air-conditioning system of the pole 1 valve hall is the same as that of the pole 2 valve hall. The drawings of the ventilation and air-conditioning system of the pole 1 valve hall refer to 30-B003701S-N0203. Drawings of ventilation and air conditioning system for pole

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

2 valve hall refer to 30-B003701S-N0204.

（1） 系统描述

（1）System description

极 1 阀厅和极 2 阀厅分别设置一套集中空调系统。系统采用一次回风变新风量全空气系统，每套系统主要由2x100%容量的组合式空气处理机组及管道系统及电控设备等组成。空气处理机组布置在相应阀厅室外地面，空调系统通过地下风道和屋架内布置的风管将处理后的空气送入阀厅内。

阀厅最小新风量为 12500m^3/h，保证室内微正压（5~10Pa）。

阀厅设独立的灾后排烟系统，换气次数不小于 0.25 次/h，排风机设置在阀厅外墙顶部。

阀厅空调系统冷热水系统为独立设置，冷冻水供回水温度为 7/12℃，热水供回水温度为 45/40℃。

The pole 1 valve hall and the pole 2 valve hall are respectively equipped with a central air conditioning system. The air system is primary return air system with variable fresh air volume. Each system is mainly composed of AHUs with 2x100% capacity, pipeline system and electronic control equipment. The AHUs is arranged outside the corresponding valve hall. The air conditioning system sends the treated air into the valve hall through the air ducts arranged in the underground air duct and the roof truss.

Minimum fresh air volume of valve hall is 12500 m^3/h, which ensures indoor micro-positive pressure (5-10Pa). The valve hall is equipped with an independent smoke exhaust system after disaster. The ventilation frequency is not less than 0.25 times/h. The exhaust fan is set on the top of the outer wall of the valve hall. The cooling and hot water system of air conditioning system in valve hall is set independently. The temperature of chilled water supply and return water is 7/12 ℃, and that of hot water supply and return water is 45/40 ℃.

（2） 系统组成

(2) Form of system

每套阀厅空调系统主要由 2x100%容量的组合式空调机组、电动三通调节水阀、电动切换/调节挡板、防火阀、管道系统及电气控制设备等组成。组合式空气处理机组由回风机段、排风新风调节段、初效过滤段、中效过滤器、冷却加热盘管段、蒸发冷却段、喷雾加湿段、辅助电加热段、送风机段、亚高效过滤段、消声段、电动风阀及必要的中间检修段等功能

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

段组成。

Each air conditioning system for valve hall is mainly composed of 2x100% capacity AHUs, electric three-way regulating water valve, electric switching/regulating baffle, fire damper, pipeline system and electrical control equipment. The AHU consists of a return air fan section, a fresh and exhaust air regulation section, a primary filter section, a medium efficiency filter, a cooling/heating coil section, an evaporative cooling section, a spray humidifying section, an auxiliary electric heating section, an air supply fan section, a sub efficient filter section, an anechoic section, an electric air damper, and a necessary intermediate maintenance section.

表 3.2-1 单个阀厅通风空调系统配置表

Table 3.2-1 Configuration table of HVAC system of single Valve Hall

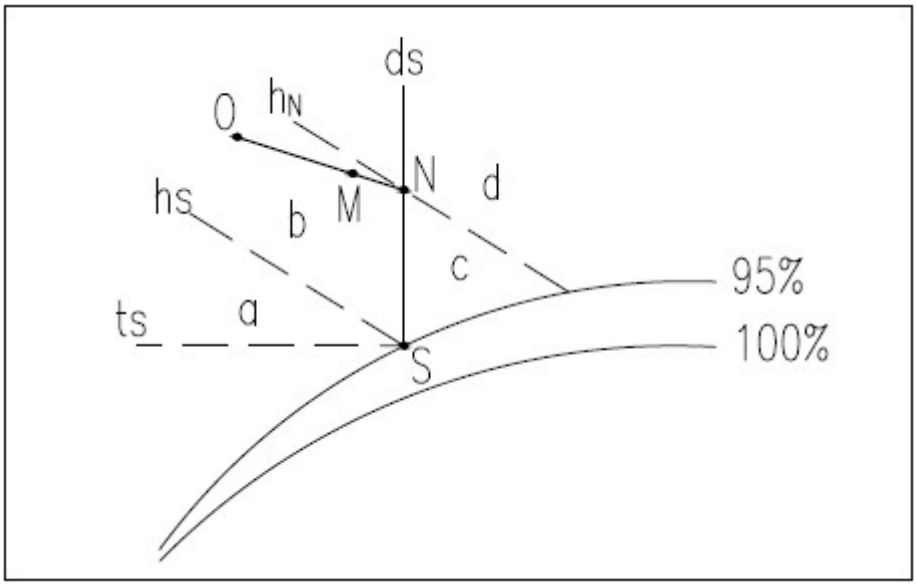
服务区域 Area	容量配置 AHU Configuration	房间送风量 Room air supply m^3/h	排风机配置 Exhaust fan configuration	备注 Remarks
极 1 阀厅 Valve Hall Pole 1 (极 2 阀厅 Valve Hall Pole 2)	2x100% 风量 Air volume：50000 m^3/h/set； 制冷量 Cooling capacity：500kW/set； 制热量 Heating capacity：550kW/set； 电加热量 Electric heating capacity：150kW/set； 加湿量 Humidifying capacity：150kg/h/set； 蒸发冷却量 Evaporating capacity：500kg/h/set	50000	2set 14580m^3/h	灾后排烟 Smoke exhaust after extinguishing

（3） 系统运行和控制

（3）Operation and Control of system

- 阀厅空调系统夏季空气处理过程如下图所示：
- Air treatment of the air conditioning system：

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任



M:新风和回风的混合点，S:送风点，O:新风状态点，N:室内状态点

M:Fresh air and return air mixing point，S:Air supply point，O:Fresh air point，N: Indoor air point

工况 a.夏季，当 $hO<hS_0$（设定值），$tO\geqslant tS_0$（设定值）时，变新风工况运行，关闭风冷螺杆式热泵机组和循环水泵，开启空气处理机内的送、回风机，开启蒸发冷却器。根据阀厅室内温度，调节新回风比，然后采用蒸发冷却器对混合后的空气进行等焓加湿处理至送风点，送风进入室内实现降温。

工况 b.夏季，当 $dO<dS_0$（设定值），$hO\geqslant hS_0$（设定值）时，全新风工况运行，开启空气处理机内的送、回风机，混风电动风阀关闭、新风电动风阀全开和排风电动风阀全开，开启风冷螺杆式热泵机组和循环水泵，开启蒸发冷却器。根据阀厅室内温度，调节电动三通调节阀开度，新风经冷却盘管降温，然后采用蒸发冷却器等焓加湿处理至送风点，送风进入室内实现降温。

工况 c.夏季，当 $dO\geqslant dS_0$（设定值），$hO<hN_0$（设定值）时，全新风工况运行，开启风冷螺杆式热泵机组和循环水泵，开启空气处理机内的送、回风机，混风电动风阀关闭、新风电动风阀全开和排风电动风阀全开。根据阀厅室内温度，调节电动三通调节阀开度，新风经冷却盘管降温去湿处理至送风点，送风进入室内实现降温。

工况 d.夏季，当 $dO\geqslant dS_0$（设定值），$hO\geqslant hN_0$（设定值）时，最小新风比工况运行，开启风冷螺杆式热泵机组和循环水泵，开启空气处理机内的送、回风机，混风电动风阀开启、新风电动风阀调节至最小新风量开度，排风电动风阀关闭。根据阀厅室内温度，调节电动三通调节阀开度，新回风混风经冷却盘管降温去湿处理至送风点，送风进入室内实现降温。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

工况 e 过渡季，换流阀设备运行时，当 tO＜tS_0（设定值）时，全新风工况运行，关闭风冷螺杆式热泵机组和循环水泵，开启空气处理机内的送、回风机，混风电动风阀关闭、新风电动风阀全开和排风电动风阀全开。根据阀厅室内相对湿度，调节加湿器运行。

工况 f.冬季，换流阀设备不运行时，最小新风比制热工况运行，开启风冷螺杆式热泵机组和循环水泵，开启空气处理机内的送、回风机，开启混风电动风阀、新风电动风阀调节至最小新风量开度，排风电动风阀关闭。根据阀厅室内温度，调节电动三通调节水阀；根据阀厅室内相对湿度，调节加湿器运行。

此外，阀厅空调系统需满足以下相关要求：当送风温度低于室内露点温度时，打开辅助电加热提高送风温度，使送风温度高于室内设计状态点露点温度 2℃。

Condition a.In Summer，when hO < hS_0 (set value) and tO < tS_0 (set value), system running under variable fresh air condition, turn off the air-cooled screw heat pump unit and circulating water pump, open the air supply and return fans in the AHU, and open the evaporative cooler. According to the indoor temperature of the valve hall, adjust the fresh and return air ratio, then running the evaporative cooler to humidify the mixed air by equal enthalpy process, and the air supply enters the room for cooling.

Condition b.In Summer，when dO < dS_0 (set value) and hO < hS_0 (set value), system running under all fresh air operation mode, open air supply and return fans in the AHU, mixer electric air damper closed, fresh air electric air damper fully opened and exhaust electric air damper fully opened, open air-cooled screw heat pump unit and circulating water pump, open evaporative cooler. According to the indoor temperature of the valve hall, the opening of the electric three-way regulating valve is adjusted, and the fresh air is cooled by the cooling coil. Then the evaporative cooler is used to humidify the supply air, and the air supply enters the room for cooling.

Condition c.In Summer，when dO≥dS_0 (setting value) and hO＜hN0 (setting value), all fresh air conditions are operated. The air-cooled screw heat pump units and circulating water pumps are opened, the supply and return fans in the AHU are opened, the mixed air electric air dampers are closed, the fresh air electric air dampers are fully opened and the exhaust electric air dampers are fully opened. According to the indoor temperature of the valve hall, the opening of the electric three-way regulating valve is adjusted. The fresh air is cooled and dehumidified by the cooling

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

coil to the air supply point, and the air supply enters the room for cooling.

Condition d.In Summer，when $dO \geqslant dS_0$ (set value) and $hO \geqslant hN_0$ (set value), the minimum fresh air ratio is operated, the air-cooled screw heat pump unit and circulating water pump are opened, the air supply and return fans in the AHU are opened, the mixed-air electric air damper is opened, the fresh air electric air damper is adjusted to the minimum fresh air volume, and the exhaust electric air damper is closed. According to the indoor temperature of the valve hall, the opening of the electric three-way regulating valve is adjusted. The new return air mixer is cooled and dehumidified by the cooling coil to the air supply point, and the air supply enters the room for cooling.

Condition e. During the transition season, when the converter valve equipment is running, when $tO < tS_0$ (set value), system runs in all fresh air condition, closes the air-cooled screw heat pump unit and circulating water pump, opens the air supply and return fans in the AHU, closes the mixed electric air damper, opens the fresh air electric air damper and the exhaust electric air damper. According to the relative humidity in the valve hall, the humidifier can be adjusted.

Condition f. In winter, when the converter valve equipment is not running, the minimum fresh air ratio heating condition runs. Open the air-cooled screw heat pump unit and circulating water pump, open the air supply and return fans in the AHU, open the mixed electric air damper and the fresh air electric air damper to adjust to the minimum fresh air openness, and close the exhaust electric air damper. According to the indoor temperature of the valve hall, adjust the electric three-way regulating water valve; according to the relative humidity of the valve hall, adjust the operation of humidifier.

In addition, the air conditioning system of valve hall should meet the following requirements: when the air supply temperature is lower than the dew point temperature in the room, turn on auxiliary electric heating to increase the air supply temperature, so that the air supply temperature is higher than the dew point temperature in the interior design state.

- 当阀厅内发生火灾时，火灾报警信号将联锁关闭空气处理机组及送、回风总管上的防火阀，以防止火灾蔓延。
- In case of fire inside valve hall, the fire alarm signal will interlock with air handling unit and fire dampers in supply/return air ducts to be turned off to prevent the fire from spreading.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

- 当阀厅内发生火灾后，经人工确认火灾已扑灭的情况下，手动打开排烟风机、电动风阀进行排烟，另外还可开启空气处理机组内的送风机将室外新风送入阀厅内加速烟气清除。如果烟温超过 280℃，则排烟风机进口处的排烟防火阀将关闭，并发出信号关闭排烟风机。
- After fire occurs, the smoke exhaust fan cannot be turned on untill the fire has been distinguished and confirmed. In addition, the supply air fan inside air conditioning unit can be used to supply outdoor fresh air to valve hall to accelerate smoke removal. If smoke temperature exceed 280℃, the smoke fire damper will be closed automatically and a signal will be sent to turn off the smoke exhaust fan.
- 阀厅空调系统全年运行，设有集中监控系统，含 PLC 的控制系统具有如下功能：对空调系统主要参数，如冷冻水温度和压力、阀厅温度和湿度、送风温度和湿度、室外温度和湿度、室内外压差等参数进行自动监测、显示和调节，对设备的运行状态进行显示、电加热器温度超限、过滤器压差超限、阀厅温湿度超限以及设备故障等进行报警，运行设备与备用设备之间的定时和故障切换，系统运行数据的收集和分析等。 阀厅空调系统也可以在设备的就地控制柜上通过手动操作方式控制设备的启停。阀厅空调监控系统可以实现全年各个工作工况的自动切换。
- A central monitoring and control system will be provided for valve hall air conditioning system, which will be running all year around. The monitoring and control system with PLC has following functions: to monitor, display and adjust the key parameters of air conditioning system, such as temperature and pressure of chilled water, temperature and humidity in valve hall, temperature and humidity of supply air outdoor temperature and humidity, pressure differential between indoor and outdoor air, to display operation status of equipment, to give alarm for temperature over limit of electric heater, pressure differential over limit of air filter, temperature and humidity over limits in valve hall, failure of equipment. to switch periodically and in the case of fault between running equipment and standby ones, to collect and analyze the operation data of air conditioning system etc. Air conditioning system can also be turned on and off manually in control cabinets nearby the equipment.

3.3 主控楼通风和空调

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

3.3 Ventilation and air conditioning of centralized control building

3.3.1 系统范围

3.3.1 System scope

主控楼通风和空调系统图纸见 30-B003701S-N0201 卷册。根据主控楼内各房间的室内温湿度要求、房间布置情况，划分为以下各空调、蒸发冷却、通风系统：

Drawings of ventilation and air conditioning systems in the main control building can refer to volume 30-B003701S-N0201. According to the indoor temperature and humidity requirements and plan layout of each room in the main control building, it is divided into the following air conditioning, evaporative cooling and ventilation systems:

（1）空调系统

- ±0.00m 换流变直流场接口设备室空调系统
- ±0.00m 直流屏室、阀冷控制保护设备室空调系统
- 6.00m 各电气房间空调系统
- 12.00m 主控室空调系统
- 6.00m 检修备品库风机盘管系统
- 12.00m 各办公室风机盘管+新风系统
- 各蓄电池室分体空调+事故排风系统

（1）Air conditioning system

- Air conditioning system for Converter Transformer Rooms on the ±0.00m floor
- Air conditioning system for DC Distribution Panel Rooms and Cooling Control Protection Equipment Rooms on the ±0.00m floor
- Air conditioning system for each electrical room on the 6.00m floor
- Air conditioning system for the Main Control Room on the 12.00m floor
- FCU(fan coil unit) system: Maintenance Workshop & Spare Parts Storage on the 6.00m floor
- FCU + Fresh air system: Each office rooms on the 12.00m floor
- Split air conditioning + Emergency ventilation system: Each Storage Battery Room

（2）蒸发冷却系统

- ±0.00m 380/220V 交流阀组配电间蒸发冷却系统

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

- ±0.00m 极 1/极 2 阀冷却设备间蒸发冷却系统

（1）Evaporative cooling system

系统类别		楼层/房间		空调面积	房间设计冷负荷	房间设计热负荷	系统冷负荷	系统热负荷
				AC area	Design cold load	Design heat load	System cold load	System heat load
System Category		Room		m²	kW	kW	kW	kW
冷冻水/热水空调系统	Chilled water / Hot water AC system	系统2	System 2					
		极1换流变直流场接口设备室	Pole 1 Converter Transformer Room	51	12.8	2.6	17.7	4.3
		极2换流变直流场接口设备室	Pole 2 Converter Transformer Room	51	12.8	2.6	17.7	4.3
		系统3	System 3					
		极1 直流屏室	Pole 1 DC Distribution Panel Room	30	7.5	3.0	12.1	4.2
		极2 直流屏室	Pole 2 DC Distribution Panel Room	30	7.5	3.0	12.1	4.2
		极1 阀冷控制保护设备室		36	10.8	1.8	17.7	3.5
		极2 阀冷控制保护设备室		36	10.8	1.8	17.7	3.5
		系统4	System 4					
		站公用控制保护设备室	Public Control Protection Equipment Room	160	35.2	8.0	47.7	12.5
		通信机房	Communication Room	160	35.2	8.0	47.7	12.5
		二次工作间	Control Protection Workshop	20	3.0	1.0	5.3	1.5
		培训室	Trainning Room	20	3.0	1.0	5.3	1.5
		极1 控制保护设备室	Pole 1 Control Protection Equipment Room	107	23.6	5.3	31.8	8.4
		极2 控制保护设备室	Pole 2 Control Protection Equipment Room	107	23.6	5.3	31.8	8.4
		暖通配电屏柜间	HVAC distribution room	23	5.1	1.2	8.9	2.0
		公用直流屏室	Public DC Distribution Panel Room	43	8.6	4.3	12.1	5.5
		系统5	System 5					
		主控室	Main Control Room	170	42.5	17.0	68.3	29.1
风机盘管系统	FCU system	检修备品库	Maintenance Workshop & Spare Parts Storage	69	9.6	6.9		
		休息室、会议室、办公室等	Retiring Room,Conference Room,Office etc.	229	36.7	23.0		
水蒸发冷却系统	Water evaporative cooling system	系统8	System 8					
		极1 380/220V交流阀组配电间	Pole 1 380/220V AC transformer room	110	27.5	/		
		极2 380/220V交流阀组配电间	Pole 2 380/220V AC transformer room	110	27.5	/		
		系统9-1	System 9-1					
		极1 阀冷却设备间	Pole 1 Cooling Installation Room	125	25	/		
		系统9-2	System 9-2					
		极2 阀冷却设备间	Pole 2 Cooling Installation Room	125	25	/		
分体空调系统	Split air conditioning system	±0.00m层直流蓄电池室(共4个)	±0.00m Storage Battery Room (total of 4)	17	2.7	1.7		
		6.00m层直流蓄电池室(共4个)	6.00m Storage Battery Room (total of 4)	20	3.2	2.0		

- Water evaporative cooling system for the 380/220V AC transformer rooms on the ±0.00m floor
- Water evaporative cooling system for the Pole 1/Pole 2 Cooling Installation Room on the ±0.00m floor respectively

（3）通风系统

- 卫生间排风系统

（3）Ventilation system

- Exhaust system: Each toilet

表 3.3-1 主控楼各空调冷热负荷

Table 3.3-1 Cooling and heating load of each AC system in Main control building

3.3.2 ±0.00m 换流变直流场接口设备室空调系统

3.3.2 Air conditioning system for Converter Transformer Rooms on the ±0.00m floor

（1）系统描述

±0.00m 极 1 换流变直流场接口设备室和极 2 换流变直流场接口设备室合设一套集中

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

空调系统。系统采用一次回风全空气系统，系统主要由 2x100%容量的组合式空调机组、管道系统及电控设备等组成。空调机组布置主控楼的±0.00m 层暖通机房内。

换流变直流场接口设备室空调系统新风比不小于 5%，维持室内微正压。另外换流变直流场接口设备室设独立的灭火后排风机，换气次数不小于 6 次/h。排风机兼作过渡季排风。

空调系统冷热水接自主控楼区域集中冷热水系统，冷冻水供回水温度为 7/12℃，热水供回水温度为 45/40℃。

(1) System description

A centralized air conditioning system is set up for the pole 1 Converter Transformer Room and pole 2 Converter Transformer Room. The system is composed of AHUs with 2x100% capacity, pipeline system and electronic control equipment. The AHUs are arranged in the (+0.00m) floor HVAC room of the main control building.

The fresh air ratio is not less than 5%, maintaining indoor micro-positive pressure. In addition, the electrical rooms are equipped with an independent exhaust fan after fire extinguishing, and the air volume of ventilation is not less than 6 times/h.

The cold and hot water of the air conditioning system is connected with the centralized cold and hot water system in the main control building area. The supply and return temperature of chilled water is 7/12 ^{o}C, and for the hot water is 45/40 ^{o}C.

（2）系统组成

空调系统主要由 2x100%容量的组合式空调机组、电动三通调节水阀、电动切换/调节挡板、防火阀、管道系统及电气控制设备等组成。组合式空气处理机组由新回风混合段、初中效过滤段、加热表冷段、加湿挡水段、电加热段、送风机段、电动风阀及必要的中间检修段等功能段组成。

(2) System Composition

The air conditioning system is mainly composed of 2x100% capacity AHUs, electric three-way regulating water valve, electric switching/regulating baffle, fire damper, pipeline system and electrical control equipment. The AHU consists of a fresh air and return air mixing section, a fresh and middle-effect filter section, a heating and cooling section, a humidifying and water-retaining section, an electric heating section, a fan section, electric dampers and a necessary intermediate maintenance section.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

表 3.3-2 换流变直流场接口设备室空调系统配置表

Table 3.3-2 Configuration table of AC system of Converter Transformer Rooms

服务区域 Area	容量配置 AHU Configuration	房间送风量 Room air supply m^3/h	排风机配置 Exhaust fan configuration	备注 Remarks
极 1 换流变直流场接口设备室 Pole 1 Converter Transformer Room	2x100% 风量 Air volume：10000 m^3/h/set； 制冷量 Cooling capacity：45kW/set； 电加热量 Electric heating capacity：10kW/set； 加湿量 Humidifying capacity：35kg/h/set；	5000	2set 4263m^3/h	过度季兼灭火后排风 The transition season air exhausting and Ventilation after extinguishing
极 2 换流变直流场接口设备室 Pole 2 Converter Transformer Room		5000	2set 4263m^3/h	

（3）系统运行和控制

各电气房间空调系统全年运行。夏季和冬季运行时，系统按最小新风工况运行，冷热水电动三通调节阀根据室内温度自动调节开度；过渡季运行时，系统按全新风直流工况运行，空气处理机组、排风机均投入运行。

空调系统配置就地自动监控系统，就地监控系统纳入暖通中央监控系统。自动监控系统对工况切换、设备启停、设备联动、室内温湿度、设备运行时间等进行自动控制及保护，并对室内温湿度、送回风温湿度、过滤器压差、设备运行状态等进行监测及报警。

空调系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。灭火后，就地手动启动灭火后排风机运行，排除室内烟气。

(3) System operation and control

The air-conditioning system for each electrical room runs all year round. In summer and winter, the system operates under the minimum fresh air condition, and the electric three-way regulating valve is automatically adjusted according to the indoor temperature. In transition season, the system operates under the full fresh air condition, and the AHUs and exhaust fans are put into operation.

The air conditioning system is equipped with an on-site automatic monitoring system, which is integrated into the central HVAC monitoring system. The automatic monitoring system automatically controls and protects the switch of working conditions, run-stop of equipment,

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

linkage of equipment, indoor temperature and humidity, operating time of equipment, and monitors and alarms indoor temperature and humidity, return air temperature and humidity, filter pressure difference and operation status of equipment.

The air conditioning system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and closing the corresponding fire damper. After extinguishing the fire, the exhaust fan operates manually to remove indoor smoke.

3.3.3 ±0.00m 直流屏室和阀冷控制保护设备室空调系统

3.3.3 Air conditioning system for DC Distribution Panel Room and Cooling Control Protection Equipment Room on the ±0.00m floor

（1）系统描述

±0.00m 极 1 直流屏室、极 1 阀冷控制保护设备室和极 2 直流屏室、极 2 阀冷控制保护设备室合设一套集中空调系统。系统采用一次回风全空气系统，系统主要由 2x100%容量的组合式空调机组、管道系统及电控设备等组成。空调机组布置在极 1 阀厅室外地面处。

换流变直流场接口设备室空调系统新风比不小于 5%，维持室内微正压。另外直流屏室、阀冷控制保护设备室分别设独立的灭火后排风机，换气次数不小于 6 次/h。排风机兼作过渡季排风。

空调系统冷热水接自主控楼区域集中冷热水系统，冷冻水供回水温度为 7/12℃，热水供回水温度为 45/40℃。

(1) System description

A centralized air conditioning system is set up for the pole 1 and pole 2 DC Distribution Panel Room and Cooling Control Protection Equipment Room. The system is composed of AHUs with 2x100% capacity, pipeline system and electronic control equipment. The AHUs are arranged outside the main control building.

The fresh air ratio is not less than 5%, maintaining indoor micro-positive pressure. In addition, the electrical rooms are equipped with an independent exhaust fan after fire extinguishing, and the air volume of ventilation is not less than 6 times/h.

The cold and hot water of the air conditioning system is connected with the centralized cold and hot water system in the main control building area. The supply and return temperature of chilled water is 7/12 ^{o}C, and for the hot water is 45/40 ^{o}C.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

（2）系统组成

空调系统主要由 2x100%容量的组合式空调机组、电动三通调节水阀、电动切换/调节挡板、防火阀、管道系统及电气控制设备等组成。组合式空气处理机组由新回风混合段、初中效过滤段、加热表冷段、加湿挡水段、电加热段、送风机段、电动风阀及必要的中间检修段等功能段组成。

(2) System Composition

The air conditioning system is mainly composed of 2x100% capacity AHUs, electric three-way regulating water valve, electric switching/regulating baffle, fire damper, pipeline system and electrical control equipment. The AHU consists of a fresh air and return air mixing section, a fresh and middle-effect filter section, a heating and cooling section, a humidifying and water-retaining section, an electric heating section, a fan section, electric dampers and a necessary intermediate maintenance section.

表 3.3-3 直流屏室和阀冷控制保护设备室空调系统配置表

Table 3.3-3 Configuration table of AC system of DC Distribution Panel Room and Cooling Control Protection Equipment Room

<table>
<tr><th>服务区域
Area</th><th>容量配置
AHU Configuration</th><th>房间送风量
Room air supply
m^3/h</th><th>排风机配置
Exhaust fan configuration</th><th>备注
Remarks</th></tr>
<tr><td>极 1 直流屏室
Pole 1 DC Distribution Panel Room</td><td rowspan="3">2x100%
风量 Air volume：17000 m^3/h/set；
制冷量 Cooling capacity：70kW/set；
电加热量 Electric heating capacity：25kW/set；
加湿量 Humidifying capacity：55kg/h/set；</td><td>3500</td><td>1set 2737m^3/h</td><td rowspan="3">过度季兼灭火后排风
The transition season air exhausting and Ventilation after extinguishing</td></tr>
<tr><td>极 2 直流屏室
Pole 2 DC Distribution Panel Room</td><td>3500</td><td>1set 2737m^3/h</td></tr>
<tr><td>极 1 阀冷控制保护设备室
Pole 1 Cooling Control Protection Equipment Room</td><td>5000</td><td>1set 4263m^3/h</td></tr>
</table>

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

极 2 阀冷控制保护设备室 Pole 2 Cooling Control Protection Equipment Room		5000	1set 4263m^3/h	

（3）系统运行和控制

各电气房间空调系统全年运行。夏季和冬季运行时，系统按最小新风工况运行，冷热水电动三通调节阀根据室内温度自动调节开度；过渡季运行时，系统按全新风直流工况运行，空气处理机组、排风机均投入运行。

空调系统配置就地自动监控系统，就地监控系统纳入暖通中央监控系统。自动监控系统对工况切换、设备启停、设备联动、室内温湿度、设备运行时间等进行自动控制及保护，并对室内温湿度、送回风温湿度、过滤器压差、设备运行状态等进行监测及报警。

空调系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。灭火后，就地手动启动灭火后排风机运行，排除室内烟气。

(3) System operation and control

The air-conditioning system for each electrical room runs all year round. In summer and winter, the system operates under the minimum fresh air condition, and the electric three-way regulating valve is automatically adjusted according to the indoor temperature. In transition season, the system operates under the full fresh air condition, and the AHUs and exhaust fans are put into operation.

The air conditioning system is equipped with an on-site automatic monitoring system, which is integrated into the central HVAC monitoring system. The automatic monitoring system automatically controls and protects the switch of working conditions, run-stop of equipment, linkage of equipment, indoor temperature and humidity, operating time of equipment, and monitors and alarms indoor temperature and humidity, return air temperature and humidity, filter pressure difference and operation status of equipment.

The air conditioning system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and closing the corresponding fire damper. After extinguishing the fire, the exhaust fan operates manually to remove indoor smoke.

3.3.4 6.00m 各电气房间空调系统

3.3.4 Air conditioning system for Electrical Equipment Rooms on the 6.00m floor

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

（1）系统描述

6.00m 站公用控制保护设备室、通信机房、二次工作间、培训室、极 1 控制保护设备室、极 2 控制保护设备室、暖通配电屏柜间、公用直流屏室合设一套集中空调系统。系统采用一次回风全空气系统，系统主要由 2x100%容量的组合式空调机组、管道系统及电控设备等组成。空调机组布置主控楼的 6.00m 层暖通机房内。

各电气设备间空调系统新风比不小于 5%，维持室内微正压。另外各电气房间分别设独立的灭火后排风机，换气次数不小于 6 次/h。

空调系统冷热水接自主控楼区域集中冷热水系统，冷冻水供回水温度为 7/12℃，热水供回水温度为 45/40℃。

(1) System description

A centralized air conditioning system is set up for the Public DC Distribution Panel Room, HVAC distribution room, Control Protection Equipment Room, Control Protection Equipment Room, Communication Room, Trainning Room and Control Protection Workshop. The system is composed of AHUs with 2x100% capacity, pipeline system and electronic control equipment. The AHUs are arranged in the (6.00m) floor HVAC room of the main control building.

The fresh air ratio is not less than 5%, maintaining indoor micro-positive pressure. In addition, the electrical rooms are equipped with an independent exhaust fan after fire extinguishing, and the air volume of ventilation is not less than 6 times/h.

The cold and hot water of the air conditioning system is connected with the centralized cold and hot water system in the main control building area. The supply and return temperature of chilled water is 7/12 ^{o}C, and for the hot water is 45/40 ^{o}C.

（2）系统组成

空调系统主要由 2x100%容量的组合式空调机组、电动三通调节水阀、电动切换/调节挡板、防火阀、管道系统及电气控制设备等组成。组合式空气处理机组由回风机段、排风段、新回风混合段、初中效过滤段、加热表冷段、加湿挡水段、电加热段、送风机段、电动风阀及必要的中间检修段等功能段组成。

(2) System Composition

The air conditioning system is mainly composed of 2x100% capacity AHUs, electric three-way regulating water valve, electric switching/regulating baffle, fire damper, pipeline

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

system and electrical control equipment. The AHU consists of a return air fan section, air mixing section, a fresh and middle-effect filter section, a heating and cooling section, a humidifying and water-retaining section, an electric heating section, a fan section, electric dampers and a necessary intermediate maintenance section.

表 3.3-4 6.00m 各电气房间空调系统配置表

Table 3.3-4 Configuration table of AC system of Electrical Equipment Rooms on the 6.00m floor

服务区域 Area	容量配置 AHU Configuration	房间送风量 Room air supply m^3/h	排风机配置 Exhaust fan configuration	备注 Remarks
公用直流屏室 Public DC Distribution Panel Room	2x100% 风量 Air volume：54000 m^3/h/set； 制冷量 Cooling capacity：225kW/set； 电加热量 Electric heating capacity：60kW/set； 加湿量 Humidifying capacity：180kg/h/set；	3500	1set 2737m^3/h	灭火后排风 Ventilation after extinguishing
暖通屏柜间 HVAC distribution room		2500	1set 1086m^3/h	
极 1 控制保护设备室 Pole 1 Control Protection Equipment Room		9000	1set 4263m^3/h	
极 2 控制保护设备室 Pole 2 Control Protection Equipment Room		9000	1set 4263m^3/h	
站公用控制保护设备室 Public Control Protection Equipment Room		13500	1set 7724m^3/h	
通信机房 Communication Room		13500	1set 7724m^3/h	

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

培训室 Trainning Room		1500	1set 1086m^3/h	
二次工作间 Control Protection Workshop		1500	/	/

（3）系统运行和控制

各电气房间空调系统全年运行。夏季和冬季运行时，系统按最小新风工况运行，冷热水电动三通调节阀根据室内温度自动调节开度；过渡季运行时，系统按变新风工况运行，空气处理机组运行。

空调系统配置就地自动监控系统，就地监控系统纳入暖通中央监控系统。自动监控系统对工况切换、设备启停、设备联动、室内温湿度、设备运行时间等进行自动控制及保护，并对室内温湿度、送回风温湿度、过滤器压差、设备运行状态等进行监测及报警。

空调系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。灭火后，就地手动启动灭火后排风机运行，排除室内烟气。

(3) System operation and control

The air-conditioning system for each electrical room runs all year round. In summer and winter, the system operates under the minimum fresh air condition, and the electric three-way regulating valve is automatically adjusted according to the indoor temperature. In transition season, the system operates under the fresh air condition, and the AHUs are put into operation.

The air conditioning system is equipped with an on-site automatic monitoring system, which is integrated into the central HVAC monitoring system. The automatic monitoring system automatically controls and protects the switch of working conditions, run-stop of equipment, linkage of equipment, indoor temperature and humidity, operating time of equipment, and monitors and alarms indoor temperature and humidity, return air temperature and humidity, filter pressure difference and operation status of equipment.

The air conditioning system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and closing the corresponding fire damper. After extinguishing the fire, the exhaust fan operates manually to remove indoor smoke.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

3.3.5 主控室空调系统

3.3.5 Air conditioning system for Main Control Room

（1）系统描述

12.00m 主控室设一套集中空调系统。系统采用一次回风全空气系统，系统主要由 2x100%容量的组合式空调机组、管道系统及电控设备等组成。空调机组布置主控楼的 17.00m 屋面处。

主控室空调系统新风比不小于 10%，维持室内微正压。

空调系统冷热水接自主控楼区域集中冷热水系统，冷冻水供回水温度为 7/12℃，热水供回水温度为 45/40℃。

(1) System description

A centralized air conditioning system is set up for the Main Control Room. The system is composed of AHUs with 2x100% capacity, pipeline system and electronic control equipment. The AHUs are arranged on the roof of the main control building.

The fresh air ratio is not less than 10%, maintaining indoor micro-positive pressure.

The cold and hot water of the air conditioning system is connected with the centralized cold and hot water system in the main control building area. The supply and return temperature of chilled water is 7/12 ^{o}C, and for the hot water is 45/40 ^{o}C.

（2）系统组成

空调系统主要由 2x100%容量的组合式空调机组、电动三通调节水阀、电动切换/调节挡板、防火阀、管道系统及电气控制设备等组成。组合式空气处理机组由回风机段、排风段、新回风混合段、初中效过滤段、加热表冷段、加湿挡水段、电加热段、送风机段、电动风阀及必要的中间检修段等功能段组成。

(2) System Composition

The air conditioning system is mainly composed of 2x100% capacity AHUs, electric three-way regulating water valve, electric switching/regulating baffle, fire damper, pipeline system and electrical control equipment. The AHU consists of a return air fan section, air mixing section, a fresh and middle-effect filter section, a heating and cooling section, a humidifying and water-retaining section, an electric heating section, a fan section, electric dampers and a necessary intermediate maintenance section.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

表 3.3-5 主控室空调系统配置表

Table 3.3-5 Configuration table of AC system of Main Control Room

服务区域 Area	容量配置 AHU Configuration	房间送风量 Room air supply m^3/h	排风机配置 Exhaust fan configuration	备注 Remarks
主控室 Main Control Room	2x100% 风量 Air volume：18000 m^3/h/set； 制冷量 Cooling capacity：85kW/set； 电加热量 Electric heating capacity：25kW/set； 加湿量 Humidifying capacity：65kg/h/set；	18000	/	/

（3）系统运行和控制

主控室空调系统全年运行。夏季和冬季运行时，系统按最小新风工况运行，冷热水电动三通调节阀根据室内温度自动调节开度；过渡季运行时，系统按变新风工况运行，空气处理机组运行。

空调系统配置就地自动监控系统，就地监控系统纳入暖通中央监控系统。自动监控系统对工况切换、设备启停、设备联动、室内温湿度、设备运行时间等进行自动控制及保护，并对室内温湿度、送回风温湿度、过滤器压差、设备运行状态等进行监测及报警。

空调系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。

(3) System operation and control

The air-conditioning system runs all year round. In summer and winter, the system operates under the minimum fresh air condition, and the electric three-way regulating valve is automatically adjusted according to the indoor temperature. In transition season, the system operates under the fresh air condition, and the AHUs are put into operation.

The air conditioning system is equipped with an on-site automatic monitoring system, which is integrated into the central HVAC monitoring system. The automatic monitoring system automatically controls and protects the switch of working conditions, run-stop of equipment, linkage of equipment, indoor temperature and humidity, operating time of equipment, and monitors and alarms indoor temperature and humidity, return air temperature and humidity, filter pressure difference and operation status of equipment.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

The air conditioning system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and closing the corresponding fire damper.

3.3.6 主控楼各办公室空调系统

3.3.6 Air conditioning system for Offices in The Main Control Building

主控楼 12.00m 层各办公室和 6.00m 检修备品库采用风机盘管的空调系统，其中 12.00m 层各办公室区域设置一套新风系统，满足新风量按≥30m^3/h.人的要求，新风机组布置于走道吊顶内。

风机盘管共 13 台，采用卧式暗装高静压型，带回风箱（后回风），标准工况下：风量 1360m^3 /h，出口静压 30Pa，制冷量 7.92kW，制热量 13.59kW。

新风机组共 1 台，风量：2000m^3 /h,制冷量：29kW，制热量：22kW。

空调系统冷热水接自主控楼区域集中冷热水系统，冷冻水供回水温度为 7/12℃，热水供回水温度为 45/40℃。

空调系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。

The fan coil units is used in the 12.00m floor offices and 6.00m maintenance spare warehouse of the main control building. A fresh air system is set up in the 12.00m floor office areas to meet the requirements of fresh air volume (≥30m^3/h person), and the fresh air unit is arranged in the ceiling of the corridor.

There are 13 fan-coil units, which adopt horizontal concealed high static pressure type with return air box. Under standard working conditions, the air volume is 1360 m^3/h, the outlet static pressure is 30 Pa, the cooling capacity is 7.92 kW, and the heating capacity is 13.59 kW. A fresh air unit, air volume: 2000 m^3/h, cooling capacity: 29 kW, heating capacity: 22 kW. The cold and hot water of the air conditioning system is connected with the centralized cold and hot water system in the main control building area. The supply and return temperature of chilled water is 7/12 ^{o}C, and for the hot water is 45/40 ^{o}C.

The air conditioning system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and the corresponding fire damper in the system closes.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

3.3.7 380/220V 交流阀组配电间蒸发冷却系统

3.3.7 Evaporative cooling system for 380/220V AC transformer rooms

（1）系统描述

±0.00m 极 1 380/220V 交流阀组配电间和极 2 380/220V 交流阀组配电间合设一套蒸发冷却系统。系统主要由 2x100%容量的蒸发冷却机组、管道系统及电控设备等组成。蒸发冷却机组布置主控楼的±0.00m 层暖通机房内。

380/220V 交流阀组配电间蒸发冷却系统为全新风直流形式，蒸发冷却机组送风，另设轴流风机排风，并维持室内微正压。蒸发冷却系统的排风机兼做灭火后排风机，灭火后排风的换气次数不小于 6 次/h。

蒸发冷却系统用水接自主控楼区域生活上水系统。

(1) System description

A centralized evaporating cooling system is set up for the Pole 1 and Pole 2 380/220V AC transformer rooms. The system is composed of evaporating cooling units with 2x100% capacity, pipeline system and electronic control equipment. The evaporating cooling units are arranged in the (±0.00m) floor HVAC room of the main control building.

The system runs with full fresh air, the fresh air supplied by the evaporating cooling units and exhaust by the fans. In addition, the exhaust fans run after fire extinguishing, and the air volume of ventilation is not less than 6 times/h.

The water for evaporating cooling is supplied from potable water in the main control building area.

（2）系统组成

蒸发冷却系统主要由 2x100%容量的蒸发冷却机组、电动切换挡板、防火阀、管道系统及电气控制设备等组成。蒸发冷却机组由初中效过滤段、蒸发冷却段、挡水段、送风机段、电动风阀及必要的中间检修段等功能段组成。

(2) System Composition

The evaporating cooling system is mainly composed of 2x100% capacity evaporating cooling units, electric switching/regulating baffle, fire damper, pipeline system and electrical control equipment. The evaporating cooling unit consists of a fresh and middle-effect filter section, a evaporating cooling section, a water-retaining section, a fan section, electric dampers

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

and a necessary intermediate maintenance section.

表 3.3-6 380/220V 交流阀组配电间蒸发冷却系统配置表

Table 3.3-6 Configuration table of EC system of 380/220V AC transformer rooms

服务区域 Area	容量配置 EC unit Configuration	房间送风量 Room air supply m^3/h	排风机配置 Exhaust fan configuration	备注 Remarks
极 1 380/220V 交流阀组配电间 Pole 1 380/220V AC transformer room	2x100% 风量 Air volume：18000 $m^3/h/set$； 加湿量 Humidifying capacity：180kg/h/set；	9000	2x100% 7724$m^3/h/set$	平时兼灭火后排风 Normal air exhausting and Ventilation after extinguishing
极 2 380/220V 交流阀组配电间 Pole 2 380/220V AC transformer room		9000	2x100% 7724$m^3/h/set$	

（3）系统运行和控制

各电气房间蒸发冷却系统夏季和过渡季运行。

蒸发冷却系统配置就地自动监控系统，就地监控系统纳入暖通中央监控系统。自动监控系统对设备启停、室内温湿度、设备运行时间等进行自动控制及保护，并对室内温湿度、送风温度、过滤器压差、设备运行状态等进行监测及报警。

蒸发冷却系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。灭火后，就地手动启动灭火后排风机运行，排除室内烟气。

(3) System operation and control

The evaporating cooling system for each electrical room runs in summer and transition season.

The system is equipped with an on-site automatic monitoring system, which is integrated into the central HVAC monitoring system. The automatic monitoring system automatically controls and protects the switch of working conditions, run-stop of equipment, linkage of equipment, indoor temperature and humidity, operating time of equipment, and monitors and alarms indoor temperature and humidity, return air temperature and humidity, filter pressure difference and operation status of equipment.

The evaporating cooling system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and closing the corresponding fire

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

damper. After extinguishing the fire, the exhaust fan operates manually to remove indoor smoke.

3.3.8 阀冷却设备间蒸发冷却系统

3.3.8 Evaporative cooling system for Cooling Installation Rooms

（1）系统描述

±0.00m 极 1 阀冷却设备间和极 1 阀冷却设备间分别设一套蒸发冷却系统。系统主要由 2x100%容量的蒸发冷却机组、管道系统及电控设备等组成。极 1 阀冷却设备间的蒸发冷却机组布置在极 1 阀厅室外地面处，极 2 阀冷却设备间的蒸发冷却机组布置在极 2 阀厅室外地面处。

阀冷却设备间蒸发冷却系统为全新风直流形式，蒸发冷却机组送风，另设轴流风机排风，并维持室内微正压。蒸发冷却系统的排风机兼做灭火后排风机，灭火后排风的换气次数不小于 6 次/h。

蒸发冷却系统用水接自主控楼区域生活上水系统。

(1) System description

A centralized evaporating cooling system is set up for the Pole 1 Cooling Installation Room and Pole 2 Cooling Installation Room respectively. One system is composed of evaporating cooling units with 2x100% capacity, pipeline system and electronic control equipment. The evaporating cooling units are arranged outside the main control building.

The system runs with full fresh air, the fresh air supplied by the evaporating cooling units and exhaust by the fans. In addition, the exhaust fans run after fire extinguishing, and the air volume of ventilation is not less than 6 times/h.

The water for evaporating cooling is supplied from potable water in the main control building area.

（2）系统组成

每套蒸发冷却系统主要由 2x100%容量的蒸发冷却机组、电动切换挡板、防火阀、管道系统及电气控制设备等组成。蒸发冷却机组由初中效过滤段、蒸发冷却段、挡水段、送风机段、电动风阀及必要的中间检修段等功能段组成。

(2) System Composition

Each evaporating cooling system is mainly composed of 2x100% capacity evaporating cooling units, electric switching/regulating baffle, fire damper, pipeline system and electrical

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

control equipment. The evaporating cooling unit consists of a fresh and middle-effect filter section, a evaporating cooling section, a water-retaining section, a fan section, electric dampers and a necessary intermediate maintenance section.

表 3.3-7 阀冷却设备间蒸发冷却系统配置表

Table 3.3-7 Configuration table of EC system of Cooling Installation Rooms

服务区域 Area	容量配置 EC unit Configuration	房间送风量 Room air supply m^3/h	排风机配置 Exhaust fan configuration	备注 Remarks
极 1 阀冷却设备间 Pole 1 Cooling Installation Room	2x100% 风量 Air volume：8000 $m^3/h/set$； 加湿量 Humidifying capacity：75kg/h/set；	8000	2x100% 7724$m^3/h/set$	平时兼灭火后排风 Normal air exhausting and Ventilation after extinguishing
极 2 阀冷却设备间 Pole 2 Cooling Installation Room	2x100% 风量 Air volume：8000 $m^3/h/set$； 加湿量 Humidifying capacity：75kg/h/set；	8000	2x100% 7724$m^3/h/set$	

（3）系统运行和控制

各电气房间蒸发冷却系统夏季和过渡季运行。

蒸发冷却系统配置就地自动监控系统，就地监控系统纳入暖通中央监控系统。自动监控系统对设备启停、室内温湿度、设备运行时间等进行自动控制及保护，并对室内温湿度、送风温度、过滤器压差、设备运行状态等进行监测及报警。

蒸发冷却系统与室内火灾探测系统联锁，当室内火灾探测系统报警，则系统停止运行，系统中相应的防火阀关闭。灭火后，就地手动启动灭火后排风机运行，排除室内烟气。

(3) System operation and control

The evaporating cooling system for each electrical room runs in summer and transition season.

The system is equipped with an on-site automatic monitoring system, which is integrated into the central HVAC monitoring system. The automatic monitoring system automatically controls and protects the switch of working conditions, run-stop of equipment, linkage of equipment, indoor temperature and humidity, operating time of equipment, and monitors and alarms indoor temperature and humidity, return air temperature and humidity, filter pressure difference and operation status of equipment.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

The evaporating cooling system interlocks with the indoor fire detection system. When the indoor fire detection system alarms, the system stops running and closing the corresponding fire damper. After extinguishing the fire, the exhaust fan operates manually to remove indoor smoke.

3.3.9 蓄电池室空调系统

3.3.9 Air conditioning system for Battery Rooms

±0.00m 层极 1 直流蓄电池室 1、极 1 直流蓄电池室 2、极 2 直流蓄电池室 1、极 2 直流蓄电池室 2 和 6.00m 层通信蓄电池室 1、通信蓄电池室 2、公用直流蓄电池室 1、公用直流蓄电池室 2，共 8 个。每个蓄电池室分别设置 1 台防腐防爆型热泵型分体壁挂式空调机。另外每个蓄电池室各设置 1 台防腐防爆型轴流事故排风机，事故通风量不小于 12 次/小时房间换气。

There are 8 batteries in ±0.00m floor, Pole 1 DC battery room 1, pole 1 DC battery room 2, pole 2 DC battery room 1, pole 2 DC battery room 2 and 6.00m layer communication battery room 1, communication battery room 2, public DC battery room 1, public DC battery room 2. Each battery room is equipped with one anti-corrosion and explosion-proof heat pump type split air conditioner. In addition, each battery room is equipped with an anti-corrosion and explosion-proof axial fan for emergency air exhaust. The ventilation rate of the emergency is not less than 12 times per hour.

表 3.3-8 蓄电池室通风空调系统配置表

Table 3.3-8 Configuration table of HVAC system of Battery Rooms

服务区域 Area	分体空调容量配置 Split AC Configuration	排风机配置 Exhaust fan configuration	备注 Remarks
各蓄电池室（共 8 个） Each Battery Room	1x100% 制冷量 Cooling capacity: 3.5kW， 制热量 Heating capacity: 3.9kW	1086 m^3/h	事故排风 Emergency Ventilation

分体空调机夏季制冷运行、过渡季冬季根据需要间歇运行。

事故排风机与室内氢气探测系统及火灾探测系统联锁，当室内氢气探测系统报警时，事故排风机启动运行，当火灾时，空调和风机均停止运行。

分体空调机、事故排风机就地控制运行。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs.

The emergency exhaust fan is interlocked with the indoor hydrogen detection system and the fire detection system. When the indoor hydrogen detection system alarms, the exhaust fan starts to operate. When the fire breaks out, the air conditioner and the fan stop running.

Split air conditioners and emergency exhaust fans are operated under local control.

3.4 #1 继电器室及 11kV 站用电室通风及空调

3.4 Ventilation and air conditioning of #1 Relay and 11kV Station Service House

#1 继电器室及 11kV 站用电室通风及空调见 30-B003701S-N0205 卷册。

#1 继电器室设置热泵型分体柜式空调，蓄电池室设置防腐防爆热泵型分体柜式空调，11kV 站用电室设置单冷型分体柜式空调。#1 继电器室和 11kV 站用电室另设灭火后排风机，换气次数不小于 6 次/h。蓄电池室另设防腐防爆型轴流事故排风机，事故通风量不小于 12 次/小时房间换气。

Ventilation and air conditioning of #1 relay room and 11kV station room refer to volume 30-B003701S-N0205.

#1 Relay room is equipped with heat pump split cabinet air conditioners, storage battery room is equipped with anti-corrosion and explosion-proof heat pump split cabinet air conditioners, and 11kV station room is equipped with single-cooling split cabinet air conditioners. # 1 relay room and 11 kV station room are equipped with exhaust fan after fire extinguishing. The ventilation volume is not less than 6 times/h. The storage battery room is equipped with an anti-corrosion and explosion-proof axial-flow exhaust fan. The ventilation rate of the emergency is not less than 12 times per hour.

表 3.4-1 #1 继电器室及 11kV 站用电室通风空调系统配置表

Table 3.4-1 Configuration table of HVAC system of #1 Relay and 11kV Station Service House

服务区域 Area	分体空调容量配置 Split AC Configuration	排风机配置 Exhaust fan configuration	备注 Remarks
#1 继电器室 #1 Relay room	共 6 台/Total 6 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 4 台/Total 4 sets 1905m^3/h/set	灭火后排风 Ventilation after extinguishing

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

11kV 站用电室 11kV Station Service room	共 4 台/Total 4 sets 制冷量 Cooling capacity: 7.5kW	共 2 台/Total 2 sets 1905m^3/h/set	
各蓄电池室（共 2 个） Each Battery room	共 2 台/Total 2 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 2 台/Total 2 sets 1086 m^3/h	事故排风 Emergency Ventilation

分体空调机夏季制冷运行、过渡季冬季根据需要间歇运行。空调机与室内火灾探测系统联锁，当室内火灾探测系统报警，则空调机停运。

蓄电池室的事故排风机与室内氢气探测系统及火灾探测系统联锁，当室内氢气探测系统报警时，事故排风机启动运行。当火灾时，空调和风机均停止运行。

灭火后排风机，手动启动灭火后排风机运行，排除室内烟气。

分体空调机、事故排风机和灭火后排风机均为就地控制运行。

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. The air conditioner is interlocked with the indoor fire detection system.

The emergency exhaust fan is interlocked with the indoor hydrogen detection system and the fire detection system. When the indoor hydrogen detection system alarms, the exhaust fan starts to operate. When the fire breaks out, the air conditioner and the fan stop running.

The exhaust fan starts after fire extinguishing manually to eliminate indoor smoke.

Split air conditioners and emergency exhaust fans are operated under local control.

3.5 #2 继电器室及计量小室通风及空调

3.5 Ventilation and air conditioning of #2 Relay and Metering House

#2 继电器室及计量小室通风及空调见 30-B003701S-N0206 卷册。

#2 继电器室和计量小室设置热泵型分体柜式空调，蓄电池室设置防腐防爆热泵型分体柜式空调。#2 继电器室和计量小室另设灭火后排风机，换气次数不小于 6 次/h。蓄电池室另设防腐防爆型轴流事故排风机，事故通风量不小于 12 次/小时房间换气。

Ventilation and air conditioning of #2 relay room and Metering House refer to volume 30-B003701S-N0206.

#2 Relay room and Metering House is equipped with heat pump split cabinet air conditioners, storage battery room is equipped with anti-corrosion and explosion-proof heat

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

pump split cabinet air conditioners. # 2 relay room and Metering House are equipped with exhaust fan after fire extinguishing. The ventilation volume is not less than 6 times/h. The storage battery room is equipped with an anti-corrosion and explosion-proof axial-flow exhaust fan. The ventilation rate of the emergency is not less than 12 times per hour.

表 3.5-1 #2 继电器室及计量小室通风及空调系统配置表

Table 3.5-1 Configuration table of HVAC system of #2 Relay and Metering House

服务区域 Area	分体空调容量配置 Split AC Configuration	排风机配置 Exhaust fan configuration	备注 Remarks
#2 继电器室 #2 Relay room	共 6 台/Total 6 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 4 台/Total 4 sets 1905m^3/h/set	灭火后排风 Ventilation after extinguishing
计量小室 Metering room	共 1 台/Total 1 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 1 台/Total 1 sets 1364m^3/h/set	
各蓄电池室（共 2 个） Each Battery room	共 2 台/Total 2 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 2 台/Total 2 sets 1086 m^3/h	事故排风 Emergency Ventilation

分体空调机夏季制冷运行、过渡季冬季根据需要间歇运行。空调机与室内火灾探测系统联锁，当室内火灾探测系统报警，则空调机停运。

蓄电池室的事故排风机与室内氢气探测系统及火灾探测系统联锁，当室内氢气探测系统报警时，事故排风机启动运行。当火灾时，空调和风机均停止运行。

灭火后排风机，手动启动灭火后排风机运行，排除室内烟气。

分体空调机、事故排风机和灭火后排风机均为就地控制运行。

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. The air conditioner is interlocked with the indoor fire detection system.

The emergency exhaust fan is interlocked with the indoor hydrogen detection system and the fire detection system. When the indoor hydrogen detection system alarms, the exhaust fan starts to operate. When the fire breaks out, the air conditioner and the fan stop running.

The exhaust fan starts after fire extinguishing manually to eliminate indoor smoke.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Split air conditioners and emergency exhaust fans are operated under local control.

3.6 #3 继电器室通风及空调

3.6 Ventilation and air conditioning of #3 Relay House

#3 继电器室通风及空调见 30-B003701S-N0209 卷册。

#3 继电器室设置热泵型分体柜式空调，蓄电池室设置防腐防爆热泵型分体柜式空调。#3 继电器室另设灭火后排风机，换气次数不小于 6 次/h。蓄电池室另设防腐防爆型轴流事故排风机，事故通风量不小于 12 次/小时房间换气。

Ventilation and air conditioning of #3 relay room refer to volume 30-B003701S-N0209.

#3 Relay room is equipped with heat pump split cabinet air conditioners, storage battery room is equipped with anti-corrosion and explosion-proof heat pump split cabinet air conditioners. # 3 relay room is equipped with exhaust fan after fire extinguishing. The ventilation volume is not less than 6 times/h. The storage battery room is equipped with an anti-corrosion and explosion-proof axial-flow exhaust fan. The ventilation rate of the emergency is not less than 12 times per hour.

表 3.6-1 #3 继电器室通风及空调系统配置表

Table 3.6-1 Configuration table of HVAC system of #3 Relay House

服务区域 Area	分体空调容量配置 Split AC Configuration	排风机配置 Exhaust fan configuration	备注 Remarks
#3 继电器室 #3 Relay room	共 6 台/Total 6 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 4 台/Total 4 sets 1905m^3/h/set	灭火后排风 Ventilation after extinguishing
各蓄电池室（共 2 个） Each Battery room	共 2 台/Total 2 sets 制冷量 Cooling capacity: 7.5kW， 制热量 Heating capacity:8.3kW	共 2 台/Total 2 sets 1086 m^3/h	事故排风 Emergency Ventilation

分体空调机夏季制冷运行、过渡季冬季根据需要间歇运行。空调机与室内火灾探测系统联锁，当室内火灾探测系统报警，则空调机停运。

蓄电池室的事故排风机与室内氢气探测系统及火灾探测系统联锁，当室内氢气探测系统报警时，事故排风机启动运行。当火灾时，空调和风机均停止运行。

灭火后排风机，手动启动灭火后排风机运行，排除室内烟气。

分体空调机、事故排风机和灭火后排风机均为就地控制运行。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. The air conditioner is interlocked with the indoor fire detection system.

The emergency exhaust fan is interlocked with the indoor hydrogen detection system and the fire detection system. When the indoor hydrogen detection system alarms, the exhaust fan starts to operate. When the fire breaks out, the air conditioner and the fan stop running.

The exhaust fan starts after fire extinguishing manually to eliminate indoor smoke.

Split air conditioners and emergency exhaust fans are operated under local control.

3.7 400V 站用电室通风及空调

3.7 Ventilation and air conditioning of 400V Station Service House

400V 站用电室通风及空调见 30-B003701S-N0208 卷册。

400V 站用电室设置单冷型分体柜式空调。400V 站用电室室另设灭火后排风机，换气次数不小于 6 次/h。

Ventilation and air conditioning of 400V Station Service House refer to volume 30-B003701S-N0208.

400V Station Service House is equipped with single-cooling split cabinet air conditioners. 400V Station Service House is equipped with exhaust fan after fire extinguishing. The ventilation volume is not less than 6 times/h.

表 3.7-1 400V 站用电室通风及空调系统配置表

Table 3.7-1 Configuration table of HVAC system of 400V Station Service House

服务区域 Area	分体空调容量配置 Split AC Configuration	排风机配置 Exhaust fan configuration	备注 Remarks
400V 站用电室 400V Station Service House	共 4 台/Total 4 sets 制冷量 Cooling capacity: 12W	共 2 台/Total 2 sets 1905m^3/h/set	灭火后排风 Ventilation after extinguishing

分体空调机夏季制冷运行、过渡季根据需要间歇运行。空调机与室内火灾探测系统联锁，当室内火灾探测系统报警，则空调机停运。

灭火后排风机，手动启动灭火后排风机运行，排除室内烟气。

分体空调机、灭火后排风机均为就地控制运行。

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. The air conditioner is interlocked with the indoor fire detection system.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

The exhaust fan starts after fire extinguishing manually to eliminate indoor smoke.

Split air conditioners and emergency exhaust fans are operated under local control.

3.8 综合泵房及消防泵房通风及空调

3.8 Ventilation and air conditioning of Pump House and Fire Pump House

综合泵房及消防泵房通风及空调见 30-B003701S-N0210 卷册。

综合泵房、消防泵房和油箱设备间分别设置一套低位防雨百叶自然进风、高位轴流风机机械排风的通风系统。各泵房通风量换气次数不小于 6 次/小时。油箱设备间通风量不小于 12 次/小时房间换气。消防泵房和油箱设备间的排风机采用防爆型风机。

综合泵房电气设备间和消防泵房电气设备间分别设置分体空调机，同时另设独立的灭火后排风机，灭火后排风量按不小于 6 次/小时计算。

Ventilation and air conditioning of Pump House and Fire Pump House refer to volume 30-B003701S-N0210.

Natural intake of low rain-proof louvers and mechanical exhaust of high-level axial-flow fans are set up for the comprehensive pump house, fire pump house and fuel tank room respectively. The ventilation rate of each pump room should not be less than 6 times per hour. The ventilation rate for fuel tank is not less than 12 times per hour. The fan in fire pump room and fuel tank room are all explosion-proof.

Electric room for pump house is equipped with split air conditioners and exhaust fan after fire extinguishing. The ventilation volume is not less than 6 times/h.

表 3.7-1 400V 站用电室通风及空调系统配置表

Table 3.7-1 Configuration table of HVAC system of 400V Station Service House

服务区域 Area	分体空调容量配置 Split AC Configuration	排风机配置 Exhaust fan configuration	备注 Remarks
综合泵房 Pump House	/	共 2 台/Total 2 sets 5581m^3/h/set	/
消防泵房 Fire Pump House	/	共 2 台/Total 2 sets 7724m^3/h/set	/

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

油箱设备间 Fuel tank equipment room	/	共 1 台/Total 1 set 1086m^3/h/set	/
综合泵房电气设备间 Pump electrical equipment room	共 1 台/Total 1 set 制冷量 Cooling capacity: 5kW， 制热量 Heating capacity:5.6kW	共 1 台/Total 1 set 1364m^3/h/set	灭火后排风 Ventilation after extinguishing
消防泵房电气设备间 Fire Pump electrical equipment room	共 1 台/Total 1 set 制冷量 Cooling capacity: 5kW， 制热量 Heating capacity:5.6kW	共 1 台/Total 1 set 1364m^3/h/set	灭火后排风 Ventilation after extinguishing

分体空调机夏季制冷运行、过渡季冬季根据需要间歇运行。

油箱设备间通风系统全年运行，综合泵房、消防泵房通风系统夏季连续运行、过渡季冬季根据需要运行。

通风空调设备与室内火灾探测系统联锁，当室内火灾探测系统报警，则空调机和风机停运。灭火后，手动启动灭火后排风机运行，排除室内烟气。

分体空调机、排风机均为就地控制运行。

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs.

Ventilation for fuel tank room runs all the year round. For pump house, the ventilation system runs in summer, and runs intermittently in transition season and winter according to needs.

The air conditioner and fans are interlocked with the indoor fire detection system. The exhaust fan starts after fire extinguishing manually to eliminate indoor smoke.

Split air conditioners and fans are operated under local control.

3.9 专用品库空调

3.9 Air conditioning of Special materials storage

专用品库空调见 30-B003701S-N0211 卷册。

专用品库设置 1 台制冷量为 5.0kW，制热量为 5.6kW 的分体空调机。

分体空调机夏季制冷运行、过渡季冬季根据需要间歇运行。分体空调机为就地控制运行。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Air conditioning of Special materials storage refer to volume 30-B003701S-N0211.

Special materials storage is equipped with 1 set of heat pump split air conditioners, the cooling capacity is 5.0 kW, and the heating capacity is 5.6 kW.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs.

Split air conditioners are operated under local control.

3.10 深井泵房通风

3.10 Ventilation of Deep well

深井泵房通风见 30-B003701S-N0212 卷册。

深井泵房设置一套低位防雨百叶自然进风、高位轴流风机机械排风的通风系统。泵房通风量换气次数不小于 6 次/小时，设置 1 台风量为 1364 m^3/h 排风机。

深井泵房通风系统夏季连续运行、过渡季冬季根据需要运行。风机为就地控制运行。

Ventilation of Deep well refer to volume 30-B003701S-N0212.

Natural intake of low rain-proof louvers and mechanical exhaust of high-level axial-flow fans are set up for Deep well. The ventilation rate should not be less than 6 times per hour, and equipped with 1 set fan of 1364 m^3/h.

Ventilation system runs in summer, and runs intermittently in transition season and winter according to needs.

3.11 站前区办公楼通风及空调

3.11 Ventilation and air conditioning of Office building in the first building area

站前区办公楼通风及空调见 30-B003701S-N0213 卷册。

站前区办公楼各办公室分别设置分体热泵型空调机，共设置 38 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，2 台制冷量为 7.5kW，制热量为 8.3kW 的分体柜式空调机，6 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

配电间设置 1 台制冷量为 5.0kW，制热量为 5.6kW 的分体空调机，另设 1 台风量为 1364 m^3/h 的灭火后排风机，满足换气次数不小于 6 次/小时。

实验室另设 1 台风量为 2737 m^3/h 的排风机，满足换气次数不小于 10 次/小时，排风机间歇运行。

站前区办公楼各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 6

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

台 300 m^3/h，3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Office building in the first building area refer to volume 30-B003701S-N0213.

Each office room is equipped with heat pump split air conditioners, and 38 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 2 sets split air conditioners (with cooling capacity of 7.5kW and heating capacity of 8.3kW), 6 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

MCC room is equipped with 1 set of split air conditioner, the cooling capacity is 5.0 kW, and the heating capacity is 5.6 kW. 1 set fan of 1364 m^3/h for ventilation after fire extinguishing is set as well, to meet the ventilation volume is not less than 6 times/h.

Laboratory is equipped with 1 set fan of 2737 m^3/h for ventilation, to meet the ventilation volume is not less than 10 times/h.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 6 sets ventilators of 300 m^3/h, 3sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.12 站前区中方宿舍通风及空调

3.12 Ventilation and air conditioning of Domitory building for chinese in the first building area

站前区中方宿舍通风及空调见 30-B003701S-N0214 卷册。

站前区中方宿舍各宿舍分别设置分体热泵型空调机，共设置 48 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机。

站前区中方宿舍各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 46 台 150 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Domitory building for chinese refer to volume

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

30-B003701S-N0214.

Each domitory room is equipped with heat pump split air conditioners, and 48 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW) are set totally.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 46 sets ventilators of 150 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.13 中方食堂通风及空调

3.13 Ventilation and air conditioning of Refectory for chinese

中方食堂通风及空调见 30-B003701S-N0215 卷册。

中方食堂各房间分别设置分体热泵型空调机，共设置 3 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，4 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

中方食堂的卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 1 台 150 m^3/h 的换气扇。

中方食堂各包房和餐厅设置换气扇排风，满足换气次数不小于 3 次/小时，共设置 2 台 300 m^3/h，3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Refectory for chinese refer to volume 30-B003701S-N0215.

Each room is equipped with heat pump split air conditioners, and 3 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 4 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 1 set ventilators of 150 m^3/h is set.

Each dining room and private room is equipped with ventilator to meet the ventilation volume is not less than 3 times/h. 2 sets ventilators of 300 m^3/h, 3sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

according to needs. Split air conditioners and fans are operated under local control.

3.14 活动室通风及空调

3.14 Ventilation and air conditioning of Activity house

活动室通风及空调见 30-B003701S-N0216 卷册。

活动室设置分体热泵型空调机，共设置 4 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

活动室的卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 1 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Activity house refer to volume 30-B003701S-N0216.

Activity house is equipped with heat pump split air conditioners, 4 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

Toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 1 set ventilators of 800 m^3/h is set.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.15 站前区巴方宿舍通风及空调

3.15 Ventilation and air conditioning of Domitory building for pakistani in the first building area

站前区巴方宿舍通风及空调见 30-B003701S-N0217 卷册。

站前区巴方宿舍各宿舍分别设置分体热泵型空调机，共设置 32 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机。

站前区巴方宿舍各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 8 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Domitory building for pakistani refer to volume 30-B003701S-N0217.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Each domitory room is equipped with heat pump split air conditioners, and 32 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW) are set totally.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 8 sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.16 巴方食堂通风及空调

3.16 Ventilation and air conditioning of Refectory for pakistani

巴方食堂通风及空调见 30-B003701S-N0218 卷册。

巴方食堂各房间分别设置分体热泵型空调机，共设置 1 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，4 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

巴方食堂的卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 1 台 150 m^3/h 的换气扇。

巴方食堂各包房和餐厅设置换气扇排风，满足换气次数不小于 3 次/小时，共设置 3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Refectory for pakistani refer to volume 30-B003701S-N0218.

Each room is equipped with heat pump split air conditioners, and 1 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 4 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 1 set ventilators of 150 m^3/h is set.

Each dining room and private room is equipped with ventilator to meet the ventilation volume is not less than 3 times/h. 3sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.17 门卫室通风及空调

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

3.17 Ventilation and air conditioning of Guard room

门卫室通风及空调见 30-B003701S-N0219 卷册。

门卫室设置分体热泵型空调机，共设置 1 台制冷量为 3.5kW，制热量为 3.9kW 的分体壁挂式空调机。

门卫室卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 1 台 150 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Guard room refer to volume 30-B003701S-N0219.

Guard room is equipped with heat pump split air conditioners, 1 set split air conditioners (with cooling capacity of 3.5kW and heating capacity of 3.9kW) is set.

Toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 1 set ventilators of 150 m^3/h is set.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.18 建维区办公楼通风及空调

3.18 Ventilation and air conditioning of Office building in the temporary building area

建维区办公楼通风及空调见 30-B003701S-N0220 卷册。

建维区办公楼各办公室分别设置分体热泵型空调机，共设置 28 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，4 台制冷量为 7.5kW，制热量为 8.3kW 的分体柜式空调机，2 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

配电间设置 1 台制冷量为 5.0kW，制热量为 5.6kW 的分体空调机，另设 1 台风量为 800 m^3/h 的灭火后排风机，满足换气次数不小于 6 次/小时。

建维区办公楼各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 2 台 300 m^3/h，3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Office building in the temporary building area refer to volume 30-B003701S-N0220.

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Each office room is equipped with heat pump split air conditioners, and 28 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 4 sets split air conditioners (with cooling capacity of 7.5kW and heating capacity of 8.3kW), 2 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

MCC room is equipped with 1 set of split air conditioner, the cooling capacity is 5.0 kW, and the heating capacity is 5.6 kW. 1 set fan of 800 m^3/h for ventilation after fire extinguishing is set as well, to meet the ventilation volume is not less than 6 times/h.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 2 sets ventilators of 300 m^3/h, 3sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.19 管理人员宿舍通风及空调

3.19 Ventilation and air conditioning of Domitory building for managers

管理人员宿舍通风及空调见 30-B003701S-N0221 卷册。

管理人员宿舍各宿舍分别设置分体热泵型空调机，东楼共设置 24 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机；西楼共设置 26 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，3 台制冷量为 7.5kW，制热量为 8.3kW 的分体柜式空调机，7 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

管理人员宿舍各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，东楼共设置 22 台 150 m^3/h 的换气扇，西楼共设置 20 台 150 m^3/h，2 台 300 m^3/h ，3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Domitory building for managers refer to volume 30-B003701S-N0221.

Each domitory room is equipped with heat pump split air conditioners, and 24 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW) are set totally for east building. 26 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 3 sets split air conditioners (with cooling capacity of 7.5kW and heating capacity of

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

8.3kW), 7 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally for west building.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 22 sets ventilators of 150 m^3/h are set totally for east building. 20 sets ventilators of 150 m^3/h, 2 sets ventilators of 300 m^3/h, 3sets ventilators of 800 m^3/h are set totally for west building.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.20 员工宿舍通风及空调

3.20 Ventilation and air conditioning of Domitory building for staff

员工宿舍通风及空调见 30-B003701S-N0222 卷册。

员工宿舍各宿舍分别设置分体热泵型空调机，共设置 70 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，5 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

员工宿舍各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 140 台 150 m^3/h，3 台 300 m^3/h ，3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Domitory building for staff refer to volume 30-B003701S-N0222.

Each domitory room is equipped with heat pump split air conditioners, and 70 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 5 sets split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 140 sets ventilators of 150 m^3/h, 3 sets ventilators of 300 m^3/h, 3 sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.21 巴方安保宿舍通风及空调

3.21 Ventilation and air conditioning of Domitory building for security

巴方安保宿舍通风及空调见 30-B003701S-N0223 卷册。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

巴方安保宿舍各宿舍分别设置分体热泵型空调机，共设置 16 台制冷量为 5.0kW，制热量为 5.6kW 的分体壁挂式空调机，1 台制冷量为 12kW，制热量为 13kW 的分体柜式空调机。

巴方安保宿舍各卫生间设置换气扇排风，满足换气次数不小于 10 次/小时，共设置 1 台 150 m^3/h，3 台 800 m^3/h 的换气扇。

空调机夏季连续运行，冬季与过渡季根据需要运行。分体空调机和排风扇均为就地手动控制运行。

Ventilation and air conditioning of Domitory building for security refer to volume 30-B003701S-N0223.

Each domitory room is equipped with heat pump split air conditioners, and 16 sets split air conditioners (with cooling capacity of 5.0kW and heating capacity of 5.6kW), 1 set split air conditioners (with cooling capacity of 12kW and heating capacity of 13kW) are set totally.

Each toilet is equipped with ventilator to meet the ventilation volume is not less than 10 times/h. 1 set ventilators of 150 m^3/h, 3 sets ventilators of 800 m^3/h are set totally.

Split air conditioner runs in summer, and runs intermittently in transition season and winter according to needs. Split air conditioners and fans are operated under local control.

3.22 防火防爆

3.22 Fire proof and explosion proof

3.22.1 通风空调系统的防火设施

3.22.1 Fire protection facilities for ventilation and air conditioning systems

各建筑物通风、空调系统均设置防火安全措施。对有人员疏散要求的内走道、防烟楼梯间、前室及面积超过规范规定要求的房间考虑设置防排烟设施。

阀厅设有灾后排烟系统。

各空调及通风系统的风管在穿越防火分区处、暖通机房的隔墙和楼板处、重要或火灾危险性大的房间隔墙和楼板处等均设置防火阀隔断，当风管内空气温度达到 70℃时，防火阀自动熔断关闭，并连锁停运空调机、通风设备。

对有气体灭火系统房间的通风空调系统，应具有灭火时设备、管道与外界隔断的功能，防止灭火气体泄漏，同时设置灭火后排风设施。

防排烟设备应符合相关标准的规定，通风空调系统中风管及保温材料应采用不燃材料。

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

Fire protection facilities are installed in ventilation and air conditioning systems of buildings. The smoke control facilities will be considered in the inner aisle, staircase, front room and areas with standard requirements.

Valve hall is set the smoke exhaust system after extinguishing .

A fire proof damper will be mounted when the duct through these areas: Fire partition, HVAC room wall and floor, important or dangerous room wall and floor and so on.When the air temperature in the duct reaches 70 ℃, the fire damper is self-contained. And air conditioners and ventilators are interlocked to shut down.

The ventilation and air conditioning system for a room with a gas fire extinguishing system, should have the function of partition, to prevent the leakage of the fire extinguishing gas, and at the same time set the exhaust fans as ventilation after fire extinguished.

Smoke control equipment should comply satisfied with the relevant standards. Ventilation and air conditioning systems should be made of non-combustible materials.

3.22.2 通风空调系统与消防控制系统联锁要求

3.22.2 Fire control interlocking requirements for ventilation and air conditioning systems

通风和空调设备与火灾探测系统的联锁控制，火灾时消防控制系统发出信号停运相应通风空调系统中的空调机组、通风机，并反馈信号至消防中心。

Ventilation and air conditioning equipment are interlocked with fire detection system. Fire control system sends stop signals to air conditioning units and ventilators, and the equipment feedback signals to fire control center.

当阀厅内发生火灾后，经人工确认火灾已扑灭的情况下，手动打开排烟风机、电动风阀进行排烟，另外还可开启空气处理机组内的送风机将室外新风送入阀厅内加速烟气清除。如果烟温超过 280℃，则排烟风机进口处的排烟防火阀将关闭，并发出信号关闭排烟风机。

After fire occurs, the smoke exhaust fan cannot be turned on untill the fire has been distinguished and confirmed. In addition, the supply air fan inside air conditioning unit can be used to supply outdoor fresh air to valve hall to accelerate smoke removal. If smoke temperature exceed 280℃, the smoke fire damper will be closed automatically and a signal will be sent to turn off the smoke exhaust fan.

3.22.3 通风空调系统防爆要求

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

3.22.3 Explosion protection requirements

在具有爆炸危险区域内的通风及空气调节设备应采用防爆式。这些区域内通风及空气调节系统上所有设备、管道采取有效的防静电措施。

可燃气体报警区域内的通风设备，应与可燃气体报警装置连锁。

Explosion-proof ventilation and air conditioning equipment shall be used for the explosive danger areas. Effective anti-static measures for all ventilation and air conditioning equipment and piping in these areas.

The ventilation equipment in the alarm area of combustible gas should be linked to the combustible gas alarm device.

4 暖通专业施工图卷册目录

4 Drawing list of HVAC Part

序号 No.	卷册检索号 Volumes Number	卷册名称 Volumes Name	备注 Remark
1	30-B003701S-N0101	暖通设计说明 Design Specification of HVAC System	
2	30-B003701S-N0102	暖通设备及材料清册 List of HVAC Equipment and Major Materials	
3	30-B003701S-N0201	主控楼通风及空调 Ventilation and Air Conditioning of Main Control Building	
4	30-B003701S-N0203	极 1 阀厅通风及空调 Ventilation and Air Conditioning of Pole 1 Valve Hall	
5	30-B003701S-N0204	极 2 阀厅通风及空调 Ventilation and Air Conditioning of Pole 2 Valve Hall	
6	30-B003701S-N0205	#1 继电器室及 11kV 站用电室通风及空调 Ventilation and Air Conditioning of #1 Relay and 11kV Station Service House	
7	30-B003701S-N0206	#2 继电器室及计量小室通风及空调 Ventilation and Air Conditioning of #2 Relay and Metering House	
8	30-B003701S-N0207	#3 继电器室通风及空调 Ventilation and air conditioning of #3 Relay House	
9	30-B003701S-N0208	132kV 继电器室通风及空调 Ventilation and air conditioning of 400V Station Service House	

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任

10	30-B003701S-N0209	400V 站用电室通风及空调 Ventilation and air conditioning of 400V Station Service House	
11	30-B003701S-N0210	综合泵房及消防泵房通风及空调 Ventilation and air conditioning of Pump House and Fire Pump House	
12	30-B003701S-N0211	专用品库通风及空调 Air conditioning of Special materials storage	
13	30-B003701S-N0212	深井泵房通风 Ventilation of Deep well	
14	30-B003701S-N0213	站前区办公楼通风及空调 Ventilation and air conditioning of Office building in the first building area	
15	30-B003701S-N0214	站前区中方宿舍通风及空调 Ventilation and air conditioning of Domitory building for chinese in the first building area	
16	30-B003701S-N0215	中方食堂通风及空调 Ventilation and air conditioning of Refectory for chinese	
17	30-B003701S-N0216	活动室通风及空调 Ventilation and air conditioning of Activity house	
18	30-B003701S-N0217	站前区巴方宿舍通风及空调 Ventilation and air conditioning of Domitory building for pakistani in the first building area	
19	30-B003701S-N0218	巴方食堂通风及空调 Ventilation and air conditioning of Refectory for pakistani	
20	30-B003701S-N0219	门卫室通风及空调 Ventilation and air conditioning of Guard room	
21	30-B003701S-N0220	建维区办公楼通风及空调 Ventilation and air conditioning of Office building in the temporary building area	
22	30-B003701S-N0221	管理人员宿舍楼通风及空调 Ventilation and air conditioning of Domitory building for managers	
23	30-B003701S-N0222	员工宿舍楼通风及空调 Ventilation and air conditioning of Domitory building for staff	
24	30-B003701S-N0223	巴方安保宿舍楼通风及空调 Ventilation and air conditioning of Domitory building for security	

本文件的知识产权为华东电力设计院有限公司所有，任何单位或个人未经许可不得复制和使用，违者将被追究法律责任